UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Computer & Information Systems Engineering

NED University of Engineering and Technology

Smart Asset Tracking and Predictive Maintenance System For Flight Catering

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Author's Declaration

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- ML Model RnD & Training
- Deployment Ready ML Model Pipeline Development
- Database Schema Definition
- Webhooks for alerts & notifications
- Middleware-DB-frontend integration

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- Research for sensors and ML models
- Widget for alerts/notifications
- Ensure accurate data forwarding
- Project flow/timeline diagrams
- View & Entry forms
- Input validation mechanisms
- Documentation & Research

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- Sensors integration with microcontroller
- Hardware assembly

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- Webhooks for Real-Time updates
- Salesforce Integration
- Tickets and alert generation logic
- Sync Swift with Salesforce
- ML Model RnD & Training
- ML Model Deployments

Executive Summary

The aviation catering industry faces significant operational challenges in managing its extensive fleets of ground support vehicles, leading to unplanned downtime, delayed services, and increased maintenance expenditures. Traditional reactive maintenance strategies and a lack of real-time asset visibility exacerbate these issues, impacting service reliability and customer satisfaction. This project, "Smart Asset Tracking and Predictive Maintenance System for Flight Catering," addresses these critical problems by developing an integrated solution that leverages the Internet of Things (IoT), Machine Learning (ML), and Enterprise Resource Planning (ERP) integration.

The system employs a custom-developed hardware unit centered around an ESP32-S3 microcontroller, equipped with GPS, DS18B20 temperature sensors, and Bluetooth Low Energy (BLE) Tyre Pressure Monitoring System (TPMS) sensors. This unit collects real-time data on vehicle location, speed, engine temperature, and tyre pressure/temperature. This data is transmitted wirelessly to the Swift Platform by Ephlux, which serves as the central data aggregation, visualization, and initial alert generation hub.

Advanced AI/ML models, specifically Bidirectional Long Short-Term Memory (Bi-LSTM) networks, are integrated to analyze the collected sensor data for predictive maintenance. These models are trained to forecast potential anomalies in engine temperature, tyre pressure, and tyre temperature, achieving high accuracy with Mean Absolute Errors (MAEs) such as 0.1795°C for engine temperature, 0.2368 MPa for tyre pressure, and 0.344°C for tyre temperature. These predictions enable proactive maintenance scheduling, minimizing unexpected breakdowns.

Further enhancing operational efficiency, the system integrates with Salesforce as an ERP solution for automated maintenance ticket management. Alerts generated by the Swift platform, triggered by either threshold breaches from hardware or AI/ML predictions, automatically create service tickets in Salesforce. Maintenance personnel can track, manage, and update these tickets within Salesforce, with status changes reflecting back to the Swift platform, ensuring a closed-loop maintenance workflow.

The system provides a user-friendly dashboard on the Swift Platform for real-time vehicle tracking, operational parameter visualization, alert notifications, and overall fleet health monitoring. By transitioning from reactive to data-driven, predictive maintenance, this project aims to significantly reduce operational costs, decrease vehicle downtime by an estimated 10-15%, improve service reliability, and enhance safety in flight catering operations. This innovative solution demonstrates a practical and scalable approach to modernizing fleet management in the demanding aviation sector.

Acknowledgments

First and foremost, we would like to express our sincere gratitude to Almighty Allah for bestowing upon us the opportunity, strength, and determination to undertake and progress through this final year project.

We are profoundly grateful to our project supervisor, ***Dr. Shahab Tehzeeb****, Assistant Professor, CIS Department, NED University of Engineering and Technology, for his invaluable guidance, unwavering support, constructive feedback, and insightful mentorship throughout the duration of this project. His expertise and encouragement were instrumental in navigating challenges and shaping the project's direction.*

We also wish to extend our special thanks to our industry collaborators at ***Ephlux****. Our sincere appreciation goes to our industry advisor,* ***Mr. Nadeem Ali****, for his continuous guidance and for meticulously ensuring that the project fulfilled all industrial requirements. His practical insights were invaluable in bridging the gap between academic concepts and real-world application. We are also deeply grateful to the CEO,* ***Mr. Ali Naseem****, for his leadership and for providing us with the essential resources and the opportunity to work with the Swift Platform, which was crucial for the project's success.*

We extend our thanks to the Department of Computer & Information Systems Engineering at NED University of Engineering and Technology for providing the academic environment and resources necessary for our research and development work.

Finally, we are thankful to our families and friends for their constant encouragement, patience, and understanding, which provided us with the moral support needed to complete this demanding undertaking.

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Abbreviations

API: Application Programming Interface

ASE: International Conference on Automated Software Engineering CSS Cascading Style Sheets

DSN: IEEE/IFIP International Conference on Dependable Systems and Networks

HTML: HyperText Markup Language

ICSE-SEIP: International Conference on Software Engineering: Software Engineering in Practice

ICWS: IEEE International Conference on Web Services

IEEE: Institute of Electrical and Electronics Engineers

JSON: JavaScript Object Notation

ML: Machine Learning

ORM: Object-Relational Mapping

Regex: Regular Expressions

SIEM: Security Information and Event Management

SOC: Security Operations Center

SQL: Structured Query Language

TDSC: IEEE Transactions on Dependable and Secure Computing

TKDE: IEEE Transactions on Knowledge and Data Engineering

TSE: IEEE Transactions on Software Engineering

UI: User Interface

UX: User Experience

XML: eXtensible Markup Language

United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace and justice. There is a total of 17 SDGs as mentioned below. Our project fulfills the following:

☐ No Poverty

☐ Zero Hunger

☐ Good Health and Well being

☐ Quality Education

☐ Gender Equality

☐ Clean Water and Sanitation

✓ Affordable and Clean Energy

✓ Decent Work and Economic Growth

✓ Industry, Innovation and Infrastructure

☐ Reduced Inequalities

✓ Sustainable Cities and Communities

✓ Responsible Consumption and Production

☐ Climate Action

☐ Life Below Water

☐ Life on Land

☐ Peace and Justice and Strong Institutions

☐ Partnerships to Achieve the Goals

Chapter 1 Introduction

1.1 Background Information

The global aviation industry, a critical component of international transport and commerce, relies heavily on a complex ecosystem of ground support services to ensure timely and efficient flight operations. Among these, aviation catering services play a pivotal role in the passenger experience, delivering meals, beverages, and other amenities to aircraft. The logistics involved in flight catering are intricate, demanding precise coordination and the reliable operation of a specialized fleet of ground vehicles, including high-loaders, service trucks, and transport vans. These vehicles operate within the demanding, fast-paced airport environment, where adherence to strict schedules is paramount.

However, the management of these catering vehicle fleets often presents significant challenges. Traditional fleet management practices in this sector have largely been reliant on manual monitoring and reactive maintenance strategies. This approach means that vehicle upkeep is typically performed only after a breakdown occurs or based on fixed, time-based schedules, which may not accurately reflect the actual condition of the assets. Consequently, aviation catering companies frequently grapple with issues such as unplanned vehicle downtime, which can lead to delayed catering deliveries to aircraft, and subsequently, flight delays. Furthermore, unexpected breakdowns often result in increased maintenance costs due to emergency repairs and the potential for more extensive damage. The lack of real-time visibility into vehicle location, status, and critical performance parameters further hampers effective fleet coordination and operational decision-making.

The operational inefficiencies stemming from these outdated practices not only inflate operational expenses but also risk diminishing client airline satisfaction and the overall passenger experience. In an industry where punctuality and service quality are key competitive differentiators, such disruptions can have significant financial and reputational repercussions.

Recognizing these persistent challenges, there is a compelling need for innovative solutions that can transform fleet management in the aviation catering sector. The advent of technologies such as the Internet of Things (IoT), advanced sensor systems, cloud computing, Artificial Intelligence (AI), and Machine Learning (ML) offers unprecedented opportunities to address these issues. By integrating these technologies, it is possible to develop intelligent systems capable of real-time asset tracking, continuous health monitoring, and predictive maintenance, thereby shifting the paradigm from reactive problem-solving to proactive, data-driven operational management. This project, "Smart

Asset Tracking and Predictive Maintenance System for Flight Catering," is conceived against this backdrop, aiming to leverage these modern technological advancements to provide a comprehensive solution tailored to the specific needs of the aviation catering industry.

1.2 Significance and Motivation

The development of a Smart Asset Tracking and Predictive Maintenance System holds considerable significance for the aviation catering industry and other fleet-dependent sectors. The core motivation behind this project stems from the pressing need to overcome the limitations of traditional fleet management and to harness the transformative potential of emerging technologies to enhance operational excellence.

Enhanced Operational Efficiency: By automating asset monitoring and enabling predictive maintenance, the system can significantly reduce vehicle downtime and optimize resource allocation. This ensures that catering services are delivered to aircraft reliably and on schedule, contributing to smoother overall airport operations.

Cost Reduction: Proactive maintenance, triggered by predictive insights, can prevent minor issues from escalating into major, costly failures. This leads to reduced expenditure on emergency repairs, spare parts, and overtime labor. Optimized fleet utilization also contributes to lower fuel consumption and operational overheads.

Improved Service Reliability and Customer Satisfaction: Timely and dependable catering services are crucial for airline satisfaction and the passenger experience. By minimizing service disruptions caused by vehicle breakdowns, the system helps catering companies meet their service level agreements and enhance their reputation.

Increased Asset Lifespan: Predictive maintenance allows for interventions at optimal times, addressing wear and tear before it leads to irreversible damage. This can extend the operational lifespan of catering vehicles, maximizing return on investment.

Enhanced Safety: Monitoring critical vehicle parameters, such as tyre pressure and engine temperature, can help identify potential safety hazards before they lead to accidents or incidents within the airport environment.

Data-Driven Decision Making: The system provides fleet managers with real-time data and actionable insights, enabling more informed decisions regarding maintenance scheduling, resource deployment, and fleet optimization.

Addressing Industry Pain Points: The project is directly motivated by the well-

documented challenges of unplanned downtime, high maintenance costs, and lack of real-time visibility within the flight catering vehicle fleets.

Leveraging Technological Advancements: The confluence of mature IoT technologies, affordable sensors, powerful cloud platforms (like the Swift Platform by Ephlux), advanced AI/ML algorithms, and robust ERP systems (like Salesforce) provides a unique opportunity to create a powerful, integrated solution.

Potential for Scalability and Adaptability: The proposed system architecture, leveraging platforms like Swift and Salesforce, is designed for scalability, allowing it to be adapted for fleets of varying sizes and potentially for other industries facing similar fleet management challenges.

Contribution to Sustainable Practices: By promoting efficient vehicle operation (e.g., optimal tyre pressure for fuel efficiency) and reducing waste associated with unexpected breakdowns and premature component replacement, the project aligns with broader goals of operational sustainability**.**

Academic and Innovative Exploration: The project offers a valuable opportunity to apply and integrate cutting-edge technologies to solve real-world industrial problems, contributing to academic knowledge and showcasing innovation in engineering.

In essence, the significance of this project lies in its potential to deliver substantial operational and economic benefits, while the motivation is driven by the desire to solve pressing industry problems through the intelligent application of modern technology.

Efficient asset tracking and maintenance are critical for industries like aviation catering, where timely operations and resource optimization are paramount.

Current fleet management practices often rely on manual monitoring and reactive maintenance, leading to inefficiencies, unexpected breakdowns, and increased operational costs. This project is motivated by the need for: Improved Operational Efficiency: Automating asset monitoring and predictive maintenance reduces vehicle downtime and optimizes resource allocation, ensuring on-time service delivery.

1.3 Problem Statement

The flight catering industry's ground support operations are frequently hampered by inefficiencies in vehicle fleet management. Key problems include:

- **Unplanned Vehicle Downtime:** Catering vehicles are susceptible to unexpected breakdowns due to a primary reliance on reactive or scheduled maintenance, rather than condition-based or predictive strategies. This downtime directly disrupts the tight schedules of airline catering, leading to service delays and potential flight schedule impacts.
- **High Maintenance Costs:** Reactive maintenance is often more expensive than planned interventions, involving emergency repair costs, overtime labor, and potentially more extensive damage to vehicles if initial warning signs are missed. Fixed-schedule maintenance can also lead to unnecessary servicing of healthy vehicles or overlook developing issues.
- **Lack of Real-Time Asset Visibility and Control:** Fleet managers often lack continuous, real-time information regarding the precise location, operational status, and critical health parameters (e.g., engine temperature, tyre pressure, fuel levels) of their vehicles. This hinders optimal dispatch, coordination, and timely response to emerging issues.
- **Inefficient Maintenance Workflow Management:** The process of identifying a fault, raising a maintenance request, scheduling repair, tracking progress, and documenting work is often manual or managed through disparate systems. This leads to delays in initiating repairs, difficulties in tracking maintenance history, and challenges in analyzing maintenance effectiveness, aspects the ERP integration aims to solve.
- **Suboptimal Resource Utilization:** Without predictive insights, scheduling of maintenance tasks and allocation of technical staff may not be optimized, leading to periods of over-utilization or under-utilization of resources.
- **Potential Safety Risks:** Undetected or unaddressed issues in vehicle health, such as critically low tyre pressure or engine overheating, can pose safety risks to personnel and other airport operations.

This project aims to address these multifaceted problems by developing an integrated system that provides real-time asset tracking, leverages AI/ML for predictive maintenance insights, and streamlines maintenance workflows through ERP integration, thereby enhancing the overall efficiency, reliability, and cost-effectiveness of flight catering fleet operations.

1.4 Aims and Objectives

The primary aim of this project is to design, develop, and implement an intelligent, integrated "Smart Asset Tracking and Predictive Maintenance System for Flight Catering" that enhances operational efficiency, reduces downtime, and optimizes maintenance processes for flight catering vehicle fleets.

To achieve this aim, the following specific objectives have been defined:

1. **Develop a Robust Hardware Integration for Real-Time Data Acquisition:**
    - ❖ Design and implement an IoT-based hardware unit using an ESP32-S3 microcontroller to interface with various sensors.
    - ❖ Integrate a NEO-6M GPS module for real-time vehicle location and speed tracking.
    - ❖ Integrate DS18B20 temperature sensors for monitoring critical temperatures (e.g., engine temperature).
    - ❖ Integrate BLE-enabled Tyre Pressure Monitoring System (TPMS) sensors to collect tyre pressure and temperature data.
    - ❖ Develop firmware for the ESP32 to collect, process, and transmit sensor data wirelessly to the Swift Platform.

2. **Implement an AI/ML-Powered Predictive Maintenance Module:**
    - ❖ Collect and preprocess time-series sensor data obtained from the hardware units via the Swift Platform.
    - ❖ Develop and train advanced machine learning models (specifically Bidirectional LSTMs) to predict potential failures or anomalies related to engine temperature, tyre pressure, and tyre temperature.
    - ❖ Achieve high predictive accuracy, targeting Mean Absolute Errors (MAEs) below 0.5 units for key parameters (e.g., 0.1795°C for engine temperature, 0.2368 MPa for tyre pressure).
    - ❖ Integrate the AI/ML models to provide predictive alerts and insights on the Swift Platform.

3. **Integrate an ERP System (Salesforce) for Automated Maintenance Management:**

   - ❖ Design a workflow for integrating alerts and predictive maintenance insights from the Swift Platform with Salesforce.
   - ❖ Automate the creation of maintenance tickets/cases in Salesforce based on critical alerts or predictions.
   - ❖ Enable maintenance personnel to manage and update the status of these tickets within Salesforce.
   - ❖ Ensure synchronization of maintenance status updates from Salesforce back to the Swift Platform for a unified operational view.

4. **Develop a Centralized Monitoring and Visualization Platform (Swift Platform):**

   - ❖ Utilize the Swift Platform by Ephlux for central data aggregation from the hardware units.
   - ❖ Design and implement a user-friendly dashboard for real-time vehicle tracking (map-based), visualization of sensor data, display of alerts (hardware-triggered and AI-predicted), and overall fleet health monitoring.

5. **Ensure System Scalability, Reliability, and Usability:**

   - ❖ Design the overall system architecture to be scalable for larger fleets and adaptable to varying operational needs.
   - ❖ Conduct thorough testing and validation of hardware components, AI/ML models, ERP integration, and the overall system to ensure reliability and accuracy.
   - ❖ Focus on creating an intuitive user experience for fleet managers and maintenance personnel.

1.5 Project Scope

The scope of this project is the successful development and demonstration of a complete, end-to-end proof-of-concept system for smart asset tracking and predictive maintenance, tailored specifically for flight catering vehicle fleets. The project delivers a fully integrated solution that validates the proposed architecture, from on-vehicle data acquisition to automated maintenance workflow management.

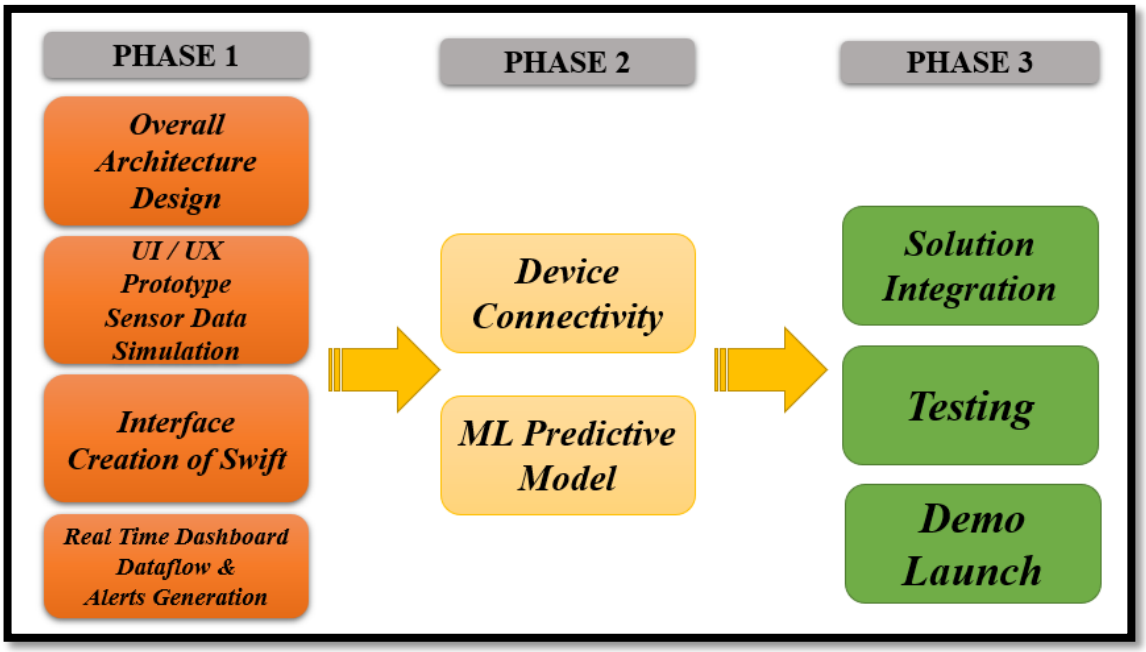


Figure 1-1 Project Phases

1. What Has Been Delivered:

The primary deliverable is a functional, multi-layered system comprising three core components.

- **Custom-built hardware prototype** has been developed, featuring an ESP32-S3 microcontroller on a custom PCB, integrated with GPS, temperature, and BLE TPMS sensors. This unit is capable of reliably collecting and transmitting vehicle telemetry.
- **Validated suite of AI/ML models** has been created; these three Bidirectional LSTM models accurately predict engine temperature, tyre pressure, and tyre temperature with proven low error rates (e.g., a Mean Absolute Error of 0.1795°C for engine temperature).

- **Integrated software ecosystem** has been established, connecting the

hardware to the Swift Platform for real-time monitoring and linking the platform's alerts to Salesforce for a closed-loop, automated maintenance ticketing process.

2. System Capabilities and Promises:

The resulting system is capable of performing several key functions. It can track the real-time location and status of vehicles on a map-based dashboard. It continuously monitors critical health parameters and can forecast potential component failures up to 24 hours in advance. Upon detecting a real-time or predicted anomaly, the system automatically generates a detailed maintenance ticket in Salesforce, ensuring no critical alert is missed. Finally, it synchronizes the status of maintenance tasks from Salesforce back to the operational dashboard, providing a unified view of both vehicle health and maintenance activity.

3. Out of Scope for Current Phase:

- Mass deployment across an entire industrial-scale fleet (the project will focus on a proof-of-concept with a limited number of units/simulated environment for some aspects).
- Development of a custom mobile application for drivers or technicians (primary interface will be the Swift Platform dashboard and Salesforce).
- Advanced security features beyond standard practices for data transmission (e.g., end-to-end hardware-level encryption beyond WiFi WPA2, though HTTPS for webhooks is implied).
- Direct control actions on the vehicle from the system (the system focuses on monitoring, prediction, and maintenance workflow automation).

1.6 Report Outline

This report details the design, development, implementation, and evaluation of the "Smart Asset Tracking and Predictive Maintenance System for Flight Catering." It is organized into the following chapters:

- **Chapter 1**: Introduction provides an overview of the project, including background information, its significance and motivation, the problem statement it addresses, the aims and objectives, the project scope, and this report outline.
- **Chapter 2:** Literature Review explores existing research and technologies

relevant to asset tracking, predictive maintenance, IoT in fleet management, AI/ML for industrial predictions, and the role of ERP systems in maintenance operations. It identifies gaps that this project aims to fill.

- **Chapter 3:** Hardware Design and Integration details the selection, design, and implementation of the on-vehicle IoT hardware, including the microcontroller, sensors, PCB, and firmware development for data acquisition and transmission.
- **Chapter 4:** AI/ML for Predictive Maintenance describes the development of the machine learning models, including data collection, preprocessing, feature engineering, model architecture (Bi-LSTMs), training, and evaluation for predicting vehicle component anomalies.
- **Chapter 5:** ERP Integration with Salesforce explains the integration of the system with Salesforce for automated maintenance ticket management, detailing the workflow, data synchronization, and Salesforce configuration.
- **Chapter 6:** System Implementation, Testing, and Overall Platform discusses the consolidated system architecture, the configuration of the Swift Platform as the central hub and UI, and the comprehensive testing methodology employed, along with results and challenges.
- **Chapter 7:** Conclusion and Future Work summarizes the project's achievements against its objectives, discusses limitations, and provides recommendations for future enhancements and research directions.
- References lists all the cited sources and relevant literature.
- Appendices may include supplementary materials such as detailed code snippets, schematics, and model configurations.

Chapter 2 Literature Review

2.1 Introduction

The development of a "Smart Asset Tracking and Predictive Maintenance System for Flight Catering" builds upon extensive research and existing technological implementations in the fields of IoT, fleet management, and machine learning. This chapter presents a critical review of several key existing systems and research papers, focusing on the specific, practical application of these technologies by other researchers to achieve goals similar to our own.

The objective of this review is to move beyond a theoretical overview and provide a detailed examination of influential works that form the foundation of modern intelligent fleet management. This involves a deep dive into specific architectural designs for IoT-based vehicle monitoring, a critical analysis of various machine learning methodologies that have been successfully applied to predictive maintenance in transportation, and an exploration of frameworks for integrating operational data with enterprise-level platforms.

By dissecting these alternate approaches, their methodologies, their reported successes, and their inherent limitations, this review will establish a clear and comprehensive context for the specific contributions and innovations of this project. It aims to demonstrate a thorough understanding of the current state-of-the-art and to precisely identify the research and implementation gaps that this project seeks to address.

2.2 Case Study: Architectures for Real-Time IoT Fleet Monitoring

The foundational layer of any intelligent fleet system is its ability to reliably track assets and collect data in real-time. The literature presents several architectural blueprints for this purpose, primarily leveraging IoT devices and cloud computing. A representative example is the **"Real-Time Fleet Management System Based on IoT and Cloud Computing"** proposed by Le et al. (2021) [1]. The stated goal of their system was to provide core fleet management functionalities, including live vehicle tracking, historical trip playback, and basic driver and vehicle management. Their system architecture was composed of three primary tiers: the in-vehicle IoT

device, a backend cloud server, and a front-end web application. The in-vehicle unit was built around a microcontroller responsible for interfacing with a GPS module to acquire location coordinates and a cellular module (GPRS/3G) for data transmission. This design choice highlights a common approach where cellular networks are used for wide-area connectivity. The backend was designed with a server application that listened for incoming data, parsed it, and stored it in a relational database. The front-end, a web application, then queried this database to display vehicle locations on a map and to render historical trip data. The primary contribution of this work is its validation of a complete, functional end-to-end system for tracking, demonstrating a practical and widely adopted architectural pattern.

Expanding on the choice of hardware, the work of Al-Faiz et al. (2021) on an **"IoT-Based Vehicle Tracking and Monitoring System Using ESP32"** is highly relevant [2]. Their research specifically focused on demonstrating the viability of the low-cost, high-performance ESP32 microcontroller as the central hub for a vehicle telematics unit. Their hardware consisted of an ESP32, a GPS module for location data, and a SIM800L module for GPRS communication. Instead of a custom backend, they utilized the ThingSpeak IoT platform, a cloud service that simplifies data aggregation and visualization. Data was transmitted from the ESP32 to a specific ThingSpeak channel, where it could be monitored. This study is significant because it confirms that an affordable and widely available microcontroller like the ESP32 possesses the necessary processing power and peripheral support for mobile IoT applications. It provides a strong justification for hardware selection in projects aiming for cost-effective scalability.

However, a limitation common to both these approaches is their primary focus on location tracking and basic monitoring. While essential, these systems do not typically incorporate a diverse suite of sensors needed for vehicle health diagnostics, such as tyre pressure monitors or discrete engine temperature sensors. As such, they provide the "tracking" component of an asset management system but lack the necessary data inputs for a sophisticated predictive maintenance engine.

2.3 Detailed Approaches in Machine Learning for Predictive Maintenance

The core innovation in modern fleet management lies in the application of machine learning to predict failures from sensor data. The literature showcases a clear preference for deep learning models due to their ability to handle the complex, non-linear, and temporal nature of vehicle telemetry.

1. RUL Prediction in the Airline Industry using Bi-LSTMs

A highly influential study in a related transportation sector was conducted by He et al. (2020), who developed a "Bidirectional LSTM-Based Deep Learning Approach for Remaining Useful Life (RUL) Prediction in the Airline Industry" [4]. This work is particularly relevant due to its focus on a safety-critical environment and its use of the same family of models as our project.

The authors' primary challenge was to accurately forecast the RUL of aircraft engine components based on time-series data from multiple sensors. They explicitly chose a Bi-LSTM architecture over standard LSTMs or traditional statistical models. Their rationale was that by processing the data sequence in both the forward (past to present) and backward (future to present) directions, the network could capture a more complete and nuanced understanding of the temporal patterns leading up to a failure event. Their model architecture consisted of multiple Bi-LSTM layers, followed by dense layers for the final RUL regression output. A key part of their methodology was the detailed feature engineering process, where they transformed raw sensor data into healthier, more informative features for the model. Their results demonstrated that the Bi-LSTM approach yielded significantly higher accuracy in RUL prediction compared to other models, providing strong empirical evidence for the effectiveness of this architecture in transportation-related prognostics.

2. Deep Learning for Predictive Maintenance in Commercial Trucking

Translating these concepts to ground-based fleets, Siahpour et al. (2022) demonstrated "A Deep Learning Approach for Predictive Maintenance in the Trucking Industry" [3]. This research addressed the challenges specific to commercial trucks, which operate under highly variable conditions (e.g., load, road quality, weather). Their methodology involved collecting large-scale telematics data and applying deep learning models to predict various types of vehicle faults.

Their work emphasized the importance of a robust data pipeline capable of handling vast amounts of noisy, real-world data. They discussed the challenges of data cleaning, normalization, and feature selection in a commercial fleet context. The study successfully showed that deep learning models could identify subtle patterns in the data that were indicative of future failures, patterns that would be missed by simpler threshold-based alerting systems. The significance of their work is that it validates the application of deep learning not just in a controlled aerospace environment but also in the more chaotic and variable domain of ground transportation, reinforcing the generalizability of this approach for diverse fleet types.

These studies, while powerful, often conclude with the validation of the model's predictive accuracy. They typically do not extend to detailing the practical, enterprise-level integration required to ensure that a generated prediction results in a managed maintenance workflow, which remains a significant implementation gap in purely academic research.

2.4 Frameworks for Integrating IoT with Enterprise Resource Planning (ERP)

A predictive alert has limited value if it does not trigger a concrete, trackable, and manageable business process. The literature recognizes this and emphasizes the need to bridge the gap between operational technology (OT), like IoT sensors and AI models, and information technology (IT), like ERP systems.

A foundational review by Mabkhot et al. (2018) provides a framework for the "Integration of IoT and ERP systems for a smarter logistics" [7]. The authors argue that the full potential of IoT is only realized when its data is seamlessly integrated with an ERP system. They identify several key benefits of this integration, including the automation of business processes, improved data accuracy across the organization, enhanced real-time decision-making, and streamlined supply chain and maintenance operations. Their proposed architectural models involve a middleware layer that acts as a bridge, translating and routing data from IoT devices to the relevant modules within the ERP system. For maintenance, this means an alert from an IoT device can automatically trigger a work order in the ERP's plant maintenance module. This research provides the strategic justification for the final, and most critical, integration step in any industrial IoT project.

Salesforce, with its Service Cloud platform, represents a modern, cloud-based system capable of fulfilling this ERP role for maintenance management. Its architecture is explicitly designed for integration. Through its comprehensive suite of APIs (e.g., the REST API for creating records [9]), external systems can programmatically interact with it. Furthermore, its internal automation tools, such as Salesforce Flow [8], allow administrators to define complex, code-free workflows that can be triggered by these external API calls. For example, a flow can be designed to listen for an incoming alert, automatically create and populate a new maintenance "Case," and assign it to the correct team based on the alert type. This capability to translate an external data event into an internal, managed business process is precisely what is required to close the loop in a predictive maintenance system.

2.5 Summary and Analysis of Existing Approaches

The review of these specific systems and research papers reveals a clear progression of capabilities in the field. The work of Le et al. [1] and Al-Faiz et al. [2] establishes a strong baseline for using custom IoT hardware for cloud-based vehicle monitoring, confirming the ESP32 as a viable and cost-effective hardware hub. These systems successfully address the challenge of real-time asset tracking.

The deep learning models proposed by He et al. [4] and Siahpour et al. [3] demonstrate that Bi-LSTMs and other neural networks are highly effective tools for accurately predicting failures from complex vehicle sensor data, far surpassing traditional methods. These studies solve the core analytical challenge of forecasting.

Finally, the conceptual framework detailed by Mabkhot et al. [7] provides the strategic blueprint for making these predictions actionable by integrating them with ERP systems. The capabilities of modern platforms like Salesforce [8, 9] show that this integration is technically feasible.

Despite the strengths of these individual approaches, a significant gap remains in combining all of these elements into a single, cohesive, and accessible solution specifically for a niche domain like flight catering. The reviewed IoT systems often lack predictive capabilities. The academic models for prediction, while accurate, are frequently not demonstrated within an end-to-end workflow that includes automated ticketing in an enterprise system. This project is designed to fill that

specific gap by creating a holistic solution that not only collects the data and predicts the failure but also automates the real-world response through deep integration with Salesforce. This end-to-end integration, from the on-vehicle sensor to the final, closed maintenance ticket, represents the primary innovation and contribution of this work.

2.6 Summary and Gap Analysis:

The literature indicates a strong trend towards leveraging technology for enhanced fleet management and predictive maintenance. Asset tracking using IoT and GPS is well-established, providing crucial real-time visibility. Predictive maintenance methodologies are evolving, with a clear shift from reactive/preventive approaches to data-driven, condition-based, and predictive strategies, significantly benefiting from AI/ML for analyzing complex sensor data. ML models, particularly LSTMs and Bi-LSTMs, have shown promise in time-series forecasting for vehicle health parameters. Furthermore, ERP systems like Salesforce offer powerful tools for managing maintenance workflows and integrating operational data.

Despite these advancements, several gaps or areas for specific application can be identified, particularly for specialized domains like flight catering:

- **Integrated End-to-End Solutions**: While individual components (tracking, basic PdM, ERPs) exist, truly integrated end-to-end systems that seamlessly combine custom IoT hardware, bespoke AI/ML models trained for specific vehicle types and operational conditions, a central visualization platform, and deep ERP workflow automation are less common, especially in readily available commercial off-the-shelf packages for niche industries. This project aims to provide such an integrated solution.
- **Tailored AI/ML for Specific Fleets:** Generic AI/ML solutions may not capture the unique failure modes or operational characteristics of specialized vehicles like those used in flight catering. This project focuses on developing and training models (tyre pressure, tyre temperature, engine temperature) specifically using data relevant to its target fleet.
- **Proactive and Automated Maintenance Workflow:** Many systems may predict failures or raise alerts, but the subsequent step of automatically initiating and managing the maintenance workflow within an enterprise-

level system like Salesforce is a key differentiator. This project's focus on closing the loop from prediction to maintenance execution via Salesforce addresses this.

- **Cost-Effective Custom Hardware:** Developing a cost-effective yet robust custom hardware solution (using ESP32, specific sensors) tailored to the data needs of the predictive models can be more advantageous than relying on generic, often more expensive, telematics devices that may not provide all the desired data points with the required frequency or precision.
- **Adaptability to Specific Platforms:** Leveraging a flexible IoT platform like Swift allows for custom dashboarding and integration capabilities that might be more adaptable than more rigid, pre-packaged fleet management software.

This project aims to address these gaps by developing a comprehensive, integrated system specifically for flight catering fleets. It combines tailored hardware for precise data acquisition, advanced AI/ML models for accurate predictions, the Swift Platform for centralized monitoring and alerting, and Salesforce integration for efficient, automated maintenance workflow management. This holistic approach is designed to deliver a more effective and targeted solution than relying on disparate or overly generic systems.

Chapter 3 Hardware Design and Integration

3.1 Introduction to the On-Vehicle Data Acquisition System

The foundation of the "Smart Asset Tracking and Predictive Maintenance System for Flight Catering" is its ability to reliably collect accurate, real-time data directly from the vehicles in operation. This crucial task is performed by a custom-designed on-vehicle data acquisition system. This hardware system is engineered to be a robust, self-contained unit capable of interfacing with multiple sensors, processing the collected data, and transmitting it wirelessly to the central Swift Platform for further analysis, AI/ML-driven prediction, and ERP integration.

The primary objectives of this hardware subsystem are:

- **Comprehensive Sensing:** To capture key vehicle parameters including precise location, speed, critical temperatures (such as engine temperature), and individual tyre pressures and temperatures.
- **Real-Time Data Processing and Transmission:** To perform initial processing of sensor data and transmit it at regular intervals to the cloud platform, ensuring timely data availability for monitoring and prediction.
- **Reliability in Harsh Environments:** To operate dependably within the demanding automotive environment of an airport, which involves vibrations, temperature fluctuations, and potential exposure to dust and moisture.
- **Local Alerting Capability:** To possess the ability to generate immediate local alerts for certain critical threshold breaches, complementing the centralized alerting system.
- **Power Efficiency:** To manage power consumption effectively for sustained operation.
- **Cost-Effectiveness and Scalability:** To utilize readily available, cost-effective components that allow for scalable deployment across a fleet of vehicles.

This chapter details the architecture, component selection, design considerations, and firmware development of this on-vehicle hardware unit, emphasizing its role as the primary data source for the entire predictive maintenance ecosystem.

3.2 System Requirements and Component Selection

The design of the on-vehicle data acquisition system was guided by a set of specific requirements derived from the project's overall aims and objectives. These requirements directly influenced the selection of the microcontroller, sensors, and communication modules. Studied the Swift API structure for GET and POST requests.

System Requirements:

- **Microcontroller Capabilities:**
  - ❖ Sufficient processing power to handle data from multiple sensors simultaneously.
  - ❖ Integrated Wi-Fi and Bluetooth Low Energy (BLE) for wireless communication and sensor interfacing.
  - ❖ Adequate memory (RAM and Flash) for firmware, data buffering, and OTA (Over-The-Air) update capability if planned.
  - ❖ Multiple General-Purpose Input/output (GPIO) pins for interfacing with various sensors and peripherals.
  - ❖ Support for common communication protocols (SPI, I2C, UART, OneWire).
  - ❖ Low power consumption modes to optimize battery life if directly battery-operated.
  - ❖ Cost-effectiveness for scalability.

3. **Sensor Requirements:**
   - ❖ **Location Tracking:** Accurate and reliable GPS module with a reasonable update rate (e.g., 1-5 Hz).
   - ❖ **Temperature Monitoring**: Precise temperature sensors for critical components like the engine, capable of operating in an automotive temperature range. An accuracy of ±1∘C for the DS18B20 is observed.
   - ❖ **Tyre Pressure and Temperature Monitoring**: Wireless TPMS sensors providing individual tyre pressure and temperature readings, with good battery life and reliable BLE communication.
   - ❖ **Durability**: Sensors must be robust enough to withstand vehicle

vibrations and environmental conditions.

4. **Data Transmission:**

   - ❖ Reliable Wi-Fi connectivity to transmit data to the Swift Platform**.**
   - ❖ Ability to format data into JSON payloads for standardized HTTP communication.

5. **Power Supply:**

   - ❖ Ability to operate from a vehicle's power supply (e.g., 12V/24V converted to 5V/3.3V) or have an independent power source with power management features.

6. **Physical Assembly:**

   - ❖ A compact and durable enclosure to protect the electronics.
   - ❖ A Printed Circuit Board (PCB) to ensure reliable interconnections between components.

Component Selection:

Based on these requirements, the following key components were selected:

4 **Microcontroller: ESP32-S3**

- ❖ The ESP32-S3 was chosen due to its dual-core Xtensa LX6 processor, ample SRAM (520 KB) and flash memory, and built-in Wi-Fi and Bluetooth 5 (LE) capabilities. This provides sufficient power for handling multiple sensors, BLE scanning for TPMS, Wi-Fi communication, and running the data processing logic. The schematic of fig. 3-1 specifies an "ESP32-S3-DEVKITC-1-N8R2". ESP32 is suitable for processing thousands of hourly data points.

5 **GPS Module: NEO-6M**

- ❖ The NEO-6M is a popular and cost-effective GPS module known for its reliability and good performance. It provides latitude, longitude, and speed data via UART communication, compatible with the ESP32. It has an update rate suitable for vehicle tracking up to 5Hz. The provided

ESP32 code interfaces with it using HardwareSerial at 9600 baud.

6 **Temperature Sensor: DS18B20**

- The DS18B20 is a digital temperature sensor that uses the OneWire protocol, requiring only a single data line for communication. It offers good accuracy (±0.5∘C over much of its range, with ±1∘C mentioned as a general specification) suitable for monitoring engine or other critical component temperatures. The provided ESP32 code uses GPIO4 for the OneWire bus. Random Nerd Tutorials also provides detailed guides for its use with ESP32.

7 **Tyre Pressure Monitoring System: BLE TPMS Sensors**

- Direct BLE TPMS sensors were selected for their ability to wirelessly transmit individual tyre pressure and temperature readings. The ESP32's BLE capabilities are used to scan for these sensors (identified by unique MAC addresses) and parse their advertisement data. This method avoids complex wiring to each wheel. The provided ESP32 code includes known MAC addresses and parsing logic for these sensors.

8 **Printed Circuit Board (PCB): Custom Design**

- A custom PCB, as depicted in the schematic in fig. 3-1, was designed to provide a stable and organized platform for mounting the ESP32, sensor interfaces, and power regulation circuitry, ensuring reliable electrical connections and a compact physical form factor.

This selection of components provides a balanced solution meeting the project's functional requirements while considering cost, availability, and ease of integration.

3.3 Hardware Architecture

The hardware architecture of the on-vehicle data acquisition system is designed around a central processing hub, the ESP32-S3 microcontroller, which orchestrates data collection from a suite of specialized sensors and manages wireless communication to the cloud. The architecture, as depicted in the schematic in fig. 3-1, ensures robust interconnections and a clear data flow path from sensing to transmission.

The core of the architecture is the ESP32-S3 module (U1). It acts as the brain of the unit, executing the firmware that controls all operations. It connects to the various peripheral sensors through its GPIO pins using different communication protocols.

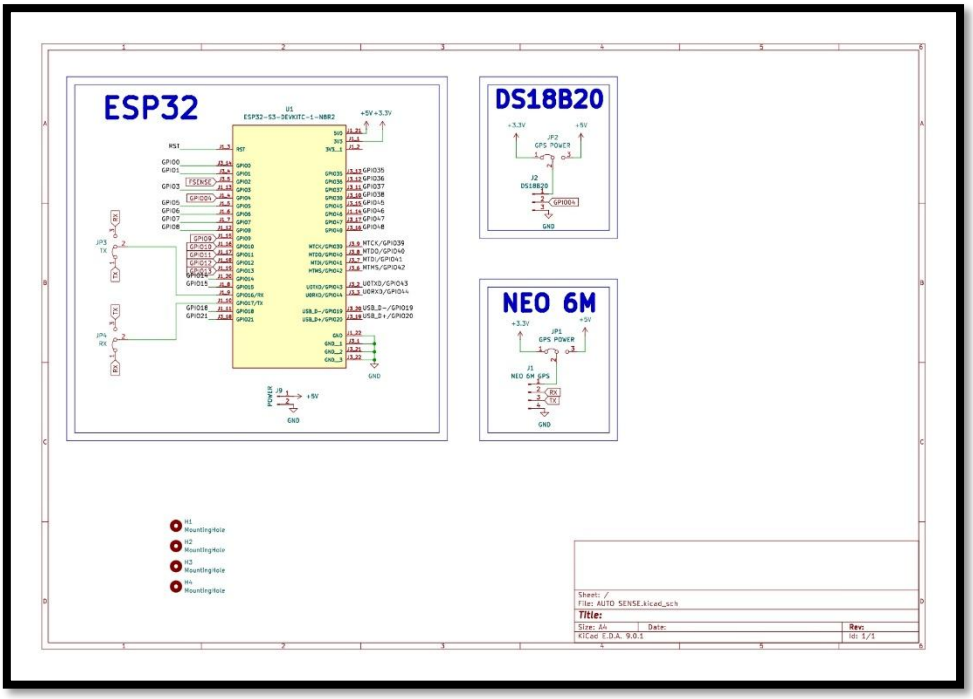


Figure 3-1 Schematic diagram of the on-vehicle data acquisition unit, detailing the interconnections between the ESP32-S3, sensors, and power circuitry.

The data flow and component interactions are as follows:

1. **Sensor Data Input:**

   - **GPS Data:** The NEO-6M GPS Module communicates with the ESP32

via a UART (serial) interface. The ESP32 continuously listens to this serial port to receive NMEA sentences containing location and speed data.

- **Engine Temperature Data:** The DS18B20 temperature sensor is connected to a single GPIO pin (GPIO4) on the ESP32 and communicates using the OneWire protocol. The ESP32 initiates a temperature reading request and receives the digital temperature value back on the same line.
- **Tyre Data (Pressure & Temperature):** The BLE TPMS sensors operate wirelessly. The ESP32's integrated BLE module is configured to actively scan for Bluetooth advertisements. When it detects an advertisement from a known TPMS sensor MAC address, it captures the manufacturer-specific data payload containing the pressure and temperature readings.
- **CAN Bus Data (Provisioned):** The schematic includes an MCP2515 CAN Bus module connected to the ESP32 via the SPI protocol (using GPIOs 9, 10, 11, 12, 13). While not utilized in the current scope of firmware for the primary predictive tasks, its inclusion in the hardware architecture provides a pathway for future expansion to read a wider array of vehicle diagnostic data directly from the vehicle's CAN bus.

2. **Central Processing:** The ESP32 firmware is responsible for parsing the raw data from each sensor into a usable format. This includes decoding NMEA sentences from the GPS, converting raw BLE data into PSI and Celsius for the TPMS, and reading the temperature from the DS18B20. It also formats this processed data into structured JSON payloads.
3. **Data Transmission:** The ESP32's integrated Wi-Fi module connects to a configured network and transmits the JSON payloads via HTTP POST requests to the predefined Swift Platform webhook URLs.
4. **Power Management**: The entire system is powered via a regulated supply that steps down the vehicle's voltage to the required 5V and 3.3V levels needed by the ESP32 and the connected sensors. This is managed on the custom PCB.

This centralized architecture allows for a modular yet integrated design, where the ESP32 efficiently manages multiple data streams and communication tasks, forming a reliable bridge between the physical vehicle and the cloud-

based analytics platform.

3.4 Key Hardware Components

This section provides a detailed description of the primary electronic components selected for the on-vehicle data acquisition unit.

1. **ESP32-S3 Microcontroller**

The ESP32-S3 is a high-performance, low-cost microcontroller that serves as the central processing unit for the hardware system. Its feature set makes it exceptionally well-suited for this project's requirements.

- **Processor:** It features a dual-core Xtensa LX6 microprocessor clocked at up to 240 MHz, providing ample processing power to manage multiple sensor inputs and communication protocols concurrently.
- **Memory:** The module is equipped with 520 KB of SRAM and, in the case of the N8R2 variant shown in the schematic, 8 MB of Flash memory, which is crucial for storing the complex firmware, buffering sensor data, and supporting potential Over-The-Air (OTA) updates.
- **Connectivity:**
  - Wi-Fi: Integrated 802.11 b/g/n Wi-Fi allows the device to connect to wireless networks and transmit data to the cloud via HTTP. The firmware leverages the WiFi.h library for this purpose.
  - **Bluetooth 5 (LE):** The integrated Bluetooth Low Energy (BLE) module is essential for interfacing with the wireless TPMS sensors. The firmware uses the BLEDevice library to actively scan for and process TPMS advertisement packets.
- **Peripherals:** The ESP32-S3 offers a rich set of peripherals, including multiple GPIOs, UARTs, SPI, I2C, and OneWire support, enabling flexible connection to a wide range of sensors like the NEO-6M GPS (UART), DS18B20 (OneWire), and the provisioned MCP2515 CAN module (SPI).

In this project, the ESP32-S3 orchestrates the timing of sensor readings, processes raw data into meaningful values, structures the data into JSON format, and manages the reliable transmission to the Swift Platform.

2. **NEO-6M GPS Module**

The NEO-6M GPS module is responsible for providing accurate real-time location and speed data for the vehicle.

- **Functionality**: It receives signals from GPS satellites to calculate its precise geographical position (latitude and longitude), altitude, and velocity.
- **Interface**: It communicates with the ESP32 via a standard serial (UART) interface, operating at a 9600 baud rate as configured in the firmware. The connection is made to the ESP32's second hardware serial port (Serial2) on pins GPIO 16 (RXD2) and GPIO 17 (TXD2).
- **Data Format**: The module outputs data in the standard NMEA format. The ESP32 firmware uses the TinyGPS++ library to parse these NMEA sentences efficiently and extract valid location and speed data.
- **Performance**: The module has a 5 Hz update rate, allowing for frequent position updates suitable for tracking a moving vehicle. The firmware ensures data validity by checking gps.location.isValid() before using or transmitting the data, filtering out readings from weak or non-existent satellite fixes.

This component is critical for the "asset tracking" aspect of the project, enabling real-time visualization on the Swift dashboard and providing contextual data for the AI/ML models.

3. **DS18B20 Temperature Sensor**

The DS18B20 is a digital temperature sensor used to monitor the engine temperature, a critical parameter for predictive maintenance.

- **Protocol**: It utilizes the OneWire communication protocol, which allows it to communicate with the ESP32 using only a single data pin (plus ground). In this project, it is connected to GPIO4 as shown in the schematic and defined in the firmware.
- **Accuracy:** The sensor offers high precision, with a general accuracy of ±0.5∘C across much of its range, and is specified with ±1∘C accuracy in the project documentation, which is sufficient for detecting overheating conditions. The AI/ML model for engine temperature

achieved a very low MAE of 0.1795°C, underscoring the quality of the data provided by this sensor.

- **Firmware Integration**: The firmware uses the DallasTemperature library, which simplifies the process of requesting and reading temperature data from the sensor over the OneWire bus.
- **Durability**: The DS18B20 is available in a waterproof, probe-style package, making it suitable for deployment in the harsh environment of a vehicle's engine bay.

This sensor provides the essential data feed for the engine temperature prediction model, enabling proactive alerts for potential overheating issues.

4. **BLE TPMS Sensors**

Wireless Tyre Pressure Monitoring System (TPMS) sensors are used to monitor the health of the vehicle's tyres, a critical factor for both safety and operational efficiency.

- **Functionality:** These are direct sensors mounted inside each tyre that measure both the internal air pressure and temperature.
- **Communication**: They operate wirelessly using Bluetooth Low Energy (BLE). They periodically broadcast their data in advertisement packets that can be picked up by a BLE receiver.
- **k**The firmware identifies the sensors by their unique MAC addresses (e.g., 80:EA:CA:13:36:92). Upon receiving a packet from a known sensor, custom parsing functions (retByte and returnData) decode the hexadecimal manufacturer data to extract pressure (converted to PSI) and temperature (in °C).
- **Update Rate:** The sensors have a typical update cycle of around 30 seconds, providing frequent updates on tyre conditions.

These sensors provide the data for two separate AI/ML models: tyre pressure prediction and tyre temperature prediction, enabling the system to forecast issues like slow leaks or tyres overheating due to under-inflation.

5. **Custom PCB**

To ensure a professional, reliable, and compact assembly of the hardware components, a custom Printed Circuit Board (PCB) was designed. The schematic for this PCB is provided in fig 3-1.

- **Purpose**: The PCB replaces fragile breadboard or wire-wrap

connections with robust copper traces, minimizing the risk of disconnections due to vehicle vibration.

- **Design**: It provides dedicated footprints for the ESP32-S3 module, connectors for the NEO-6M GPS and DS18B20 sensors, and the circuitry for the MCP2515 CAN module. It also includes power regulation circuitry to step down the vehicle's voltage to the 3.3V and 5V required by the components.
- **Connectors and Headers**: The board uses standard connectors (J-series) and headers (JP-series) for modular connection of external sensors and for power configuration, as seen in the schematic.
- **Robustness**: The design includes mounting holes (H1, H2, H3, H4) for secure installation within an enclosure, further protecting it from the operational environment.

The use of a custom PCB is a critical step.

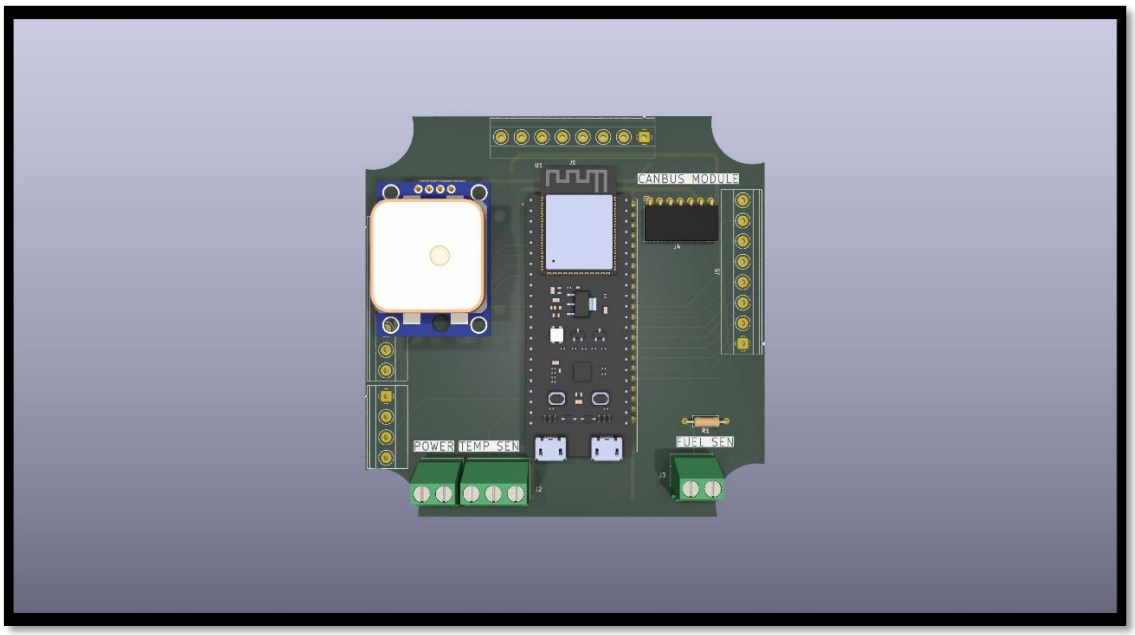


Figure 3-2 3d model of PCB generated by the step file of ki

3.5 Hardware Assembly

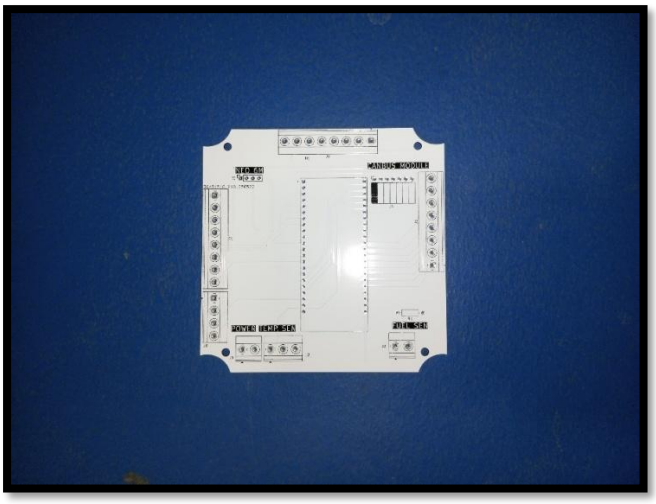


Figure 3-3 The custom-designed Printed Circuit Board (PCB) after fabrication, showing the footprints for the ESP32-S3, sensor modules, and power components



Figure 3-4 The fully assembled hardware unit, showing the ESP32-S3 and NEO-6M GPS module mounted on the custom PCB.

3.6 Final Prototype and Enclosure

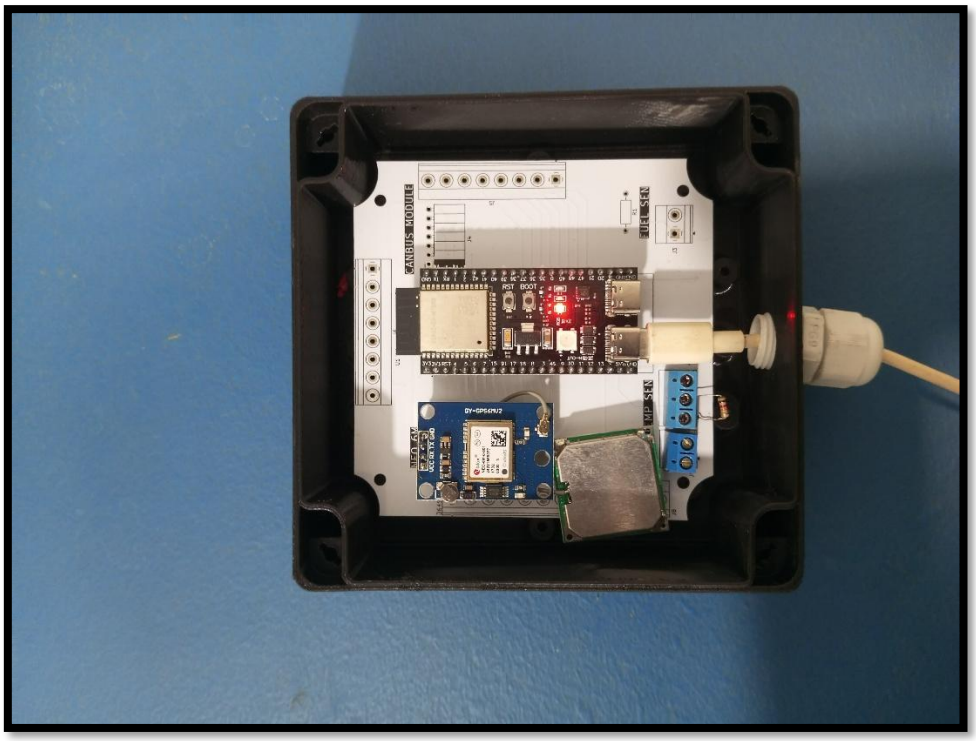


Figure 3-5 The assembled PCB fitted inside the custom 3D-printed enclosure to protect it from the operational environment.



Figure 3-6 The final hardware prototype in its protective enclosure, ready for deployment in a vehicle.

3.7 Firmware Development for ESP32

The firmware developed for the ESP32-S3 microcontroller is the software that brings the hardware components to life, enabling them to perform their

designated tasks. Written in C/C++ using the Arduino framework, the firmware is responsible for orchestrating all on-vehicle operations, from sensor data acquisition to wireless transmission and local alert generation. The following subsections detail the key aspects of the firmware's design and functionality, based on the provided source code.

4 Sensor Data Acquisition and Processing

The firmware is designed to handle data from three different types of sensors concurrently, each requiring a specific protocol and parsing logic.

- **GPS Data (NEO-6M):** The firmware initializes a hardware serial port (Serial2) at a 9600 baud rate to communicate with the GPS module. In the main loop, it continuously reads available bytes from this serial port and feeds them into a TinyGPS++ object. This library parses the raw NMEA sentences and provides structured access to location (latitude and longitude) and speed (km/h) data. The code includes a validity check (gps.location.isValid()) to ensure that only data from a valid satellite fix is processed and transmitted.
- **Engine Temperature (DS18B20**): The firmware utilizes the OneWire and DallasTemperature libraries to interface with the DS18B20 sensor connected to GPIO4. It initiates a sensors.requestTemperatures() command to prompt the sensor to take a reading and then retrieves the value in Celsius using sensors.getTempCByIndex(0). This process is executed at regular intervals within the main loop.
- **Tyre Pressure and Temperature (BLE TPMS):** To acquire data from the wireless TPMS sensors, the firmware uses the ESP32's built-in BLE functionality. It initializes a BLE scanner (BLEDevice::getScan()) and registers a custom callback handler (MyAdvertisedDeviceCallbacks). This scanner actively looks for BLE advertisement packets. The callback handler checks if the packet's MAC address matches one of the known TPMS sensor addresses. If a match is found, it extracts the hexadecimal manufacturer data payload and uses custom parsing functions (retByte, returnData) to decode the raw values into pressure (in PSI) and temperature (in °C) for the corresponding tyre.

5 Network Connectivity (Wifi, NTP)

Reliable network connectivity is crucial for transmitting data to the cloud.

- **Wi-Fi Management**: The firmware includes a robust Wi-Fi connection manager. It stores a list of known network credentials (SSID and password). Upon startup, it scans for available networks and iterates through its known list, attempting to connect to the first available one. This multi-network approach enhances the reliability of the device, allowing it to connect to different access points if the primary one is unavailable. The WiFi.h library is used to manage these functions.
- **Time Synchronization**: To ensure accurate timestamps for all data records, the firmware synchronizes its internal clock with a Network Time Protocol (NTP) server upon successful Wi-Fi connection. It is configured to use standard NTP servers like pool.ntp.org. This provides a universal time reference, which is essential for accurate time-series analysis in the AI/ML models. The provided firmware uses configTime to set up the NTP client.

6 Data Transmission to Swift Platform (JSON Payloads, HTTP POST)

Once sensor data is collected and processed, it must be transmitted to the Swift Platform.

- **JSON Formatting**: The firmware uses the ArduinoJson library to dynamically construct JSON payloads for each sensor reading. This ensures the data is sent in a structured, machine-readable format that the Swift Platform's API can easily interpret. Separate JSON objects are created for GPS data, TPMS data, engine temperature, and system alerts. Each payload includes essential information like a unique reading_id, vehicle_id, sensor_id, a precise timestamp, and a nested value object containing the actual sensor readings.
- **HTTP Communication:** Data transmission is handled via HTTP POST requests to specific webhook URLs for the Swift Platform. The firmware uses the HTTPClient library to manage these requests. It sets the Content-Type header to application/json and sends the serialized JSON string as the request body. The firmware is configured with distinct webhook URLs for posting sensor readings (POST_SENSOR_READING_WEBHOOK) and for sending alerts (POST_ALERT_WEBHOOK), ensuring data is routed correctly within the Swift Platform.

7 Local Alerting Logic

In addition to sending raw data for cloud-based analysis, the firmware has a built-in local alerting mechanism to react immediately to critical conditions.

- **Threshold-Based Alerts:** The firmware continuously checks the latest sensor readings against predefined critical thresholds. For example, it checks if tyre pressure is too high (>32 PSI) or too low (<=25 PSI), or if the engine temperature is too high (>=80°C).
- **Alert Generation:** If a threshold is breached, the firmware constructs a specific JSON payload for an alert. This payload includes details like the alert type, severity, a description of the issue, and the ID of the reading that triggered it. This alert payload is then sent immediately to the dedicated alert webhook on the Swift Platform. This ensures that critical warnings are flagged and recorded without delay.

- **Vehicle Status Update:** The firmware also tracks the number of active alerts for a given cycle. Based on this count, it determines an overall vehicle status ("Active", "Warning", or "Critical") and sends this status to a separate webhook, providing a high-level summary of the vehicle's health.

3.8 Power Management Strategy

Efficient power management is essential for the long-term, reliable operation of an in-vehicle electronic device. The hardware and firmware design incorporate several strategies to optimize power consumption.

ESP32's low-power modes can be used to reduce consumption during idle periods, extending battery life. Although the provided code runs in a continuous loop, the architecture is designed to be compatible with deep-sleep cycles where the device could wake periodically to take readings, significantly reducing power draw.

Furthermore, the sensors are "duty-cycled," meaning they are not operated continuously at their maximum capacity. The firmware logic polls the sensors at specific intervals (e.g., every 5-30 seconds) rather than constantly. For instance, the main sensor reading and transmission block is executed every 5 seconds, and the BLE scan is limited to a 1-second duration within each cycle. This duty-cycling strategy is a key aspect of managing power, ensuring that

components are only active when necessary, thus conserving energy. The system is designed to be powered by the vehicle's electrical system, with on-board voltage regulators on the PCB providing the stable 3.3V and 5V supplies required by the electronic components.

3.9 Challenges in Hardware Integration

The integration of a multi-component hardware system for a real-world automotive environment presents several challenges that were considered during the design and development process:

- **Signal Integrity and Noise**: The vehicle environment is electronically noisy. Ensuring clean signals for sensitive communications like the OneWire protocol and high-speed UART can be challenging. The use of a custom PCB with proper grounding and trace routing is a key measure to mitigate this risk.

- **Wireless Communication Reliability (Wi-Fi & BLE):** The metal body of a vehicle can interfere with wireless signals. The BLE scanning for TPMS sensors and the Wi-Fi connection for data transmission can be affected by this. This requires careful placement of the unit and its antennas within the vehicle and robust connection-handling logic in the firmware.
- **Sensor Accuracy and Calibration:** Ensuring sensors provide accurate data over time and across different operating temperatures is critical. The DS18B20 temperature sensor, for example, requires proper calibration against a reference to ensure its readings are accurate for the predictive models.
- **Firmware Complexity:** The firmware must handle multiple tasks concurrently: polling various sensors using different protocols, managing network connectivity, handling BLE scanning, and constructing/sending data payloads. Managing these tasks without blocking operations or introducing race conditions requires careful programming, making use of non-blocking timing techniques (like millis()) and structured code.
- **Physical Durability**: The hardware unit must withstand constant vibrations and a wide range of temperatures. The choice of robust components and the use of a secure enclosure and a custom PCB are

essential to ensure the physical longevity of the device.

3.10 Summary of Hardware System

The hardware system developed for this project serves as the vital sensory input for the entire predictive maintenance ecosystem. It is a well-integrated, custom-designed IoT unit centered around the powerful and versatile ESP32-S3 microcontroller. The system successfully combines a suite of carefully selected sensors—NEO-6M for GPS tracking, DS18B20 for temperature monitoring, and wireless BLE TPMS for tyre health—to capture a comprehensive set of real-time vehicle data.

The system's architecture is consolidated on a custom PCB, ensuring electrical reliability and physical robustness suitable for an automotive environment. The sophisticated firmware developed for the ESP32 acts as the system's brain, efficiently managing concurrent data acquisition from multiple sensors, processing the raw data into a structured JSON format, and transmitting it reliably via Wi-Fi to the Swift Platform. Furthermore, the firmware incorporates local intelligence to generate immediate alerts for critical safety thresholds. This hardware system effectively bridges the physical world of the vehicle with the digital world of cloud computing and data analytics, providing the high-quality, high-frequency data stream that is foundational to the project's predictive maintenance and operational monitoring objectives.

Chapter 4 AI/ML for Predictive Maintenance

4.1 Introduction to AI/ML Subsystem

The AI/ML subsystem is the predictive engine at the heart of the "Smart Asset Tracking and Predictive Maintenance System." While the hardware system provides the raw data, this module transforms that data into actionable intelligence. The primary goal of this subsystem is to move beyond simple threshold-based alerts and enable true predictive maintenance by forecasting the future state of critical vehicle components. By proactively identifying impending failures, the system allows for scheduled interventions, minimizing costly unplanned downtime and enhancing the operational reliability of the flight catering fleet.

This subsystem specifically targets three critical vehicle health indicators: tyre pressure, tyre temperature, and engine temperature. Anomalies in these parameters are often leading indicators of significant mechanical issues, such as slow leaks, tyre structural failures, or engine overheating. To achieve high predictive accuracy, the system employs state-of-the-art deep learning models, specifically Bidirectional Long Short-Term Memory (Bi-LSTM) networks. These models are trained on extensive time-series datasets collected from the vehicle sensors to learn complex temporal patterns and dependencies. The development pipeline for this subsystem encompasses a comprehensive

process, including data acquisition from the Swift Platform, rigorous data preprocessing, advanced feature engineering, model architecture design, training, and thorough evaluation using historical data to ensure robustness and reliability. This chapter details each stage of this process, from data handling to the final deployment of the predictive models.

4.2 Data Acquisition from Swift Platform

The AI/ML subsystem relies on a consistent and reliable stream of data, which is provided by the hardware units via the Swift Platform. The on-vehicle hardware, as detailed in Chapter 3, collects high-frequency sensor data and transmits it to Swift Platform webhooks. This centralized platform then serves as the primary data source for the machine learning pipeline.

The datasets used for the development and training of the predictive models were sourced from the platform's aggregated data logs. These included:

- **tyre_pressure_2024.csv:** Containing historical time-series data for tyre pressure.
- **tyre_temp_2024.csv:** Containing historical time-series data for tyre temperature.
- **vehicle_engine_temperature_UAE.csv:** Containing historical time-series data for engine temperature.

These datasets consist of thousands of hourly data points (ranging from 2,921 to over 5,000 per set), providing a rich foundation for training the deep learning models. The data, indexed by timestamps, was confirmed to be free of missing values at the source, ensuring a high-quality starting point for preprocessing. This structured data collection process is fundamental, as the quality and quantity of the input data directly influence the performance and accuracy of the resulting predictive models.

4.3 Data Preprocessing and Feature Engineering

Raw sensor data is rarely suitable for direct input into machine learning models. To prepare the data and enhance the models' ability to learn meaningful patterns, a meticulous data preprocessing and feature engineering pipeline was implemented.

1. **Data Cleaning and Normalization**

The first step in preparing the data involves cleaning and standardizing it to ensure consistency and suitability for model training.

- **Timestamp Standardization:** Timestamps across all datasets were standardized to a uniform format. This is critical for time-series analysis to ensure that data points are correctly ordered and that temporal features can be accurately extracted.
- **Data Splitting**: The aggregated dataset was strategically split to facilitate focused training and robust evaluation. A significant portion (80%) was allocated to engine temperature data, reflecting its high criticality, with 10% for other parameters and 10% reserved for validation.
- **Normalization:** The MinMaxScaler technique from the scikit-learn library was applied to normalize all numerical features to a common scale (typically between 0 and 1). This is a crucial step for neural networks as it prevents features with larger magnitudes from disproportionately influencing the model's learning process and helps improve the speed and stability of training.
- **Sequence Creation:** LSTM models require input data to be structured as sequences. A sliding window approach was used to transform the time-series data into input sequences of 40-56 hourly data points. Each sequence of historical data is mapped to a future target value that the model learns to predict (e.g., the temperature 24 hours ahead).
- **Handling NaNs:** Any Not-a-Number (NaN) values that might arise during feature engineering (e.g., from rolling window calculations at the start of a dataset) are dropped to maintain data integrity before being fed into the model.

2. **Feature Selection and Creation**

Feature engineering is the process of using domain knowledge to create new input features from the raw data, which can significantly improve model performance. The following advanced techniques were employed:

- **Temporal Lags:** Historical values of the target variables (e.g., temperature and pressure from 24 hours prior) were included as features. These 24-hour lag values provide the model with crucial historical context, allowing it to learn from recent trends and patterns.
- **Rolling Features:** Rolling statistics, such as 72-hour moving averages

and variances, were calculated for the sensor data. These features help to smooth out short-term noise and fluctuations, making underlying trends more apparent to the model. They are also effective in detecting gradual drifts or sudden spikes that might indicate an anomaly.

- **Cyclical Encodings:** Time-based features like the hour of the day, day of the week, and month of the year exhibit cyclical patterns (e.g., hour 23 is followed by hour 0. To represent this continuity to the model, sine and cosine transformations were applied (e.g., hour_sin = sin(2π * Hour / 24) and hour_cos = cos(2π * Hour / 24)). This technique allows the model to effectively learn periodic patterns, such as diurnal temperature oscillations.

- **Binary Flags:** Simple binary flags were created to represent specific operational contexts, such as weekends or rush hours. This provides the model with contextual information about the vehicle's operational state (e.g., reduced load on weekends, higher stress during rush hour), which can improve accuracy by an estimated 5-10%.

These engineered features, totaling 11 distinct inputs, provide the Bi-LSTM models with a rich, context-aware representation of the vehicle's state, which is fundamental to their high predictive accuracy.

4.4 Predictive Model Architecture (Bi-LSTMs)

To accurately forecast the vehicle health parameters, this project employs a sophisticated deep learning architecture based on Bidirectional Long Short-Term Memory (Bi-LSTM) networks. The core architecture is consistent across the three predictive models (tyre pressure, tyre temperature, and engine temperature), but each is trained independently on its respective dataset to specialize in its prediction task. The architecture is implemented using the Keras library within TensorFlow.

The general architecture of the models is as follows:

- **Input Layer:** The models accept input sequences of the 11 engineered features described in the previous section. The sequence length is typically between 40 and 56 hourly data points.
- **Bi-LSTM Layers:** The core of the models consists of two stacked Bi-LSTM layers. The first layer has 128 units, and the second has 64 units.

By processing the data in both forward and backward directions, these layers can capture complex temporal dependencies and context from the entire input sequence, which is crucial for making accurate predictions.

- **Batch Normalization:** Following each Bi-LSTM layer, a Batch Normalization layer is applied. This technique normalizes the activations of the previous layer, which helps to stabilize the learning process, accelerate training convergence, and can act as a regularizer.
- **Dropout:** A Dropout layer with a rate between 0.2 and 0.3 is included after each Batch Normalization layer. Dropout is a powerful regularization technique that randomly sets a fraction of neuron activations to zero during training. This prevents the model from becoming overly reliant on any single neuron, thus reducing overfitting and improving its ability to generalize to new, unseen data.
- **Output Layer:** A final Dense (fully connected) layer is used to produce the output predictions. The number of neurons in this layer corresponds to the desired prediction horizon.

1. **Tyre Pressure Prediction Model**

   This model is trained specifically on the tyre_pressure_2024.csv dataset. Its Dense output layer is configured with 8 units to forecast tyre pressure for the next 8 hours. The final trained model is saved as *Final_TP_Prediction_Model.h5.*

2. **Tyre Temperature Prediction Model**

   This model is trained on the tyre_temp_2024.csv dataset. Similar to the pressure model, its Dense output layer has 8 units to predict tyre temperature for the next 8 hours. The final trained model is saved as *Final_Tyre_Temp_Prediction_Model.h5.*

3. **Engine Temperature Prediction Model**

   Given the higher criticality of engine health, this model has a longer prediction horizon. It is trained on the vehicle_engine_temperature_UAE.csv dataset, and its Dense output layer is configured with 24 units, enabling it to forecast the engine temperature for the next 24 hours. The final trained model is saved as

Final_Temp_Prediction_Model.h5.

4.5 Model Evaluation and Performance Metrics

The performance of the trained models was rigorously evaluated to ensure their accuracy and reliability. The primary metric used for evaluation was the Mean Absolute Error (MAE), which measures the average magnitude of the errors in a set of predictions, without considering their direction. It provides an easily interpretable measure of prediction error in the original units of the data (e.g., °C for temperature, MPa for pressure). The target for all models was to achieve a MAE below 0.5 units.

The models consistently achieved MAEs well below this target, demonstrating exceptional performance:

- **Engine Temperature Model:** Achieved the lowest MAE of 0.1795°C. This high level of precision is critical for a parameter like engine temperature where small deviations can be significant.
- **Tyre Pressure Model**: Achieved a test MAE of 0.2368 MPa.
- **Tyre Temperature Model**: Achieved a test MAE of 0.344°C.

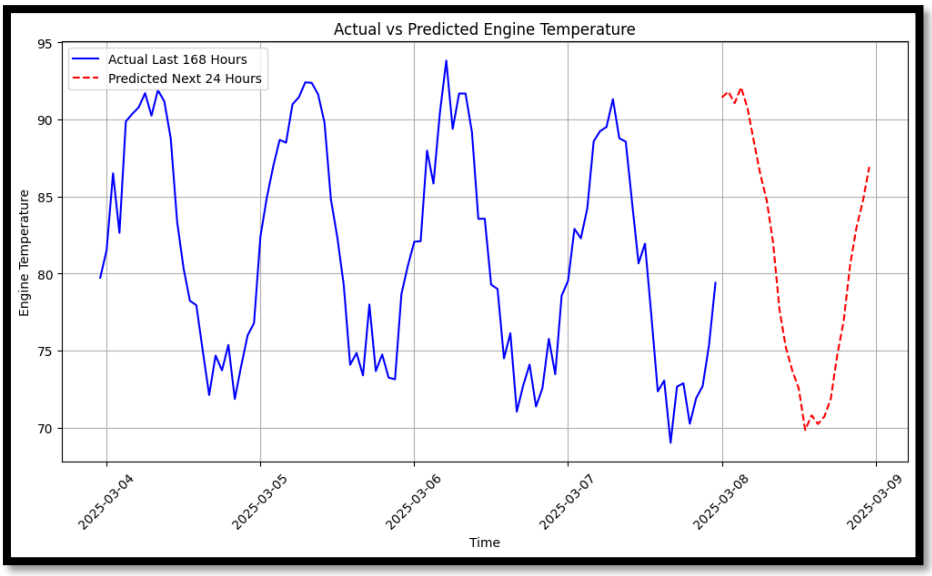


Figure 4-124-Hour Engine Temperature Forecast. The model uses the actual data from the last 168 hours (7 days) to generate a prediction for the next 24 hours.

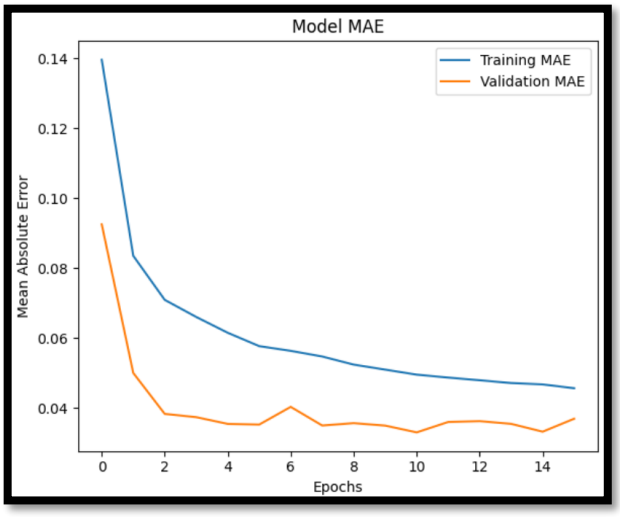


Figure 4-2 Model Training and Validation MAE Over Epochs. The graph shows the Mean Absolute Error for both the training data (blue) and the validation data (orange) decreasing as the model is trained over 15 epochs.

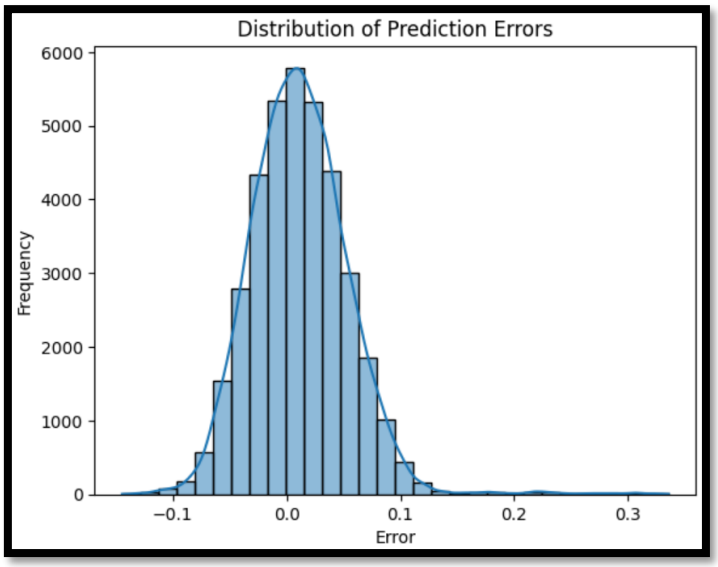


Figure 4-3 Model Training and Validation MAE Over Epochs. The graph shows the Mean Absolute Error for both the training data (blue) and the validation data (orange) decreasing as the model is trained.

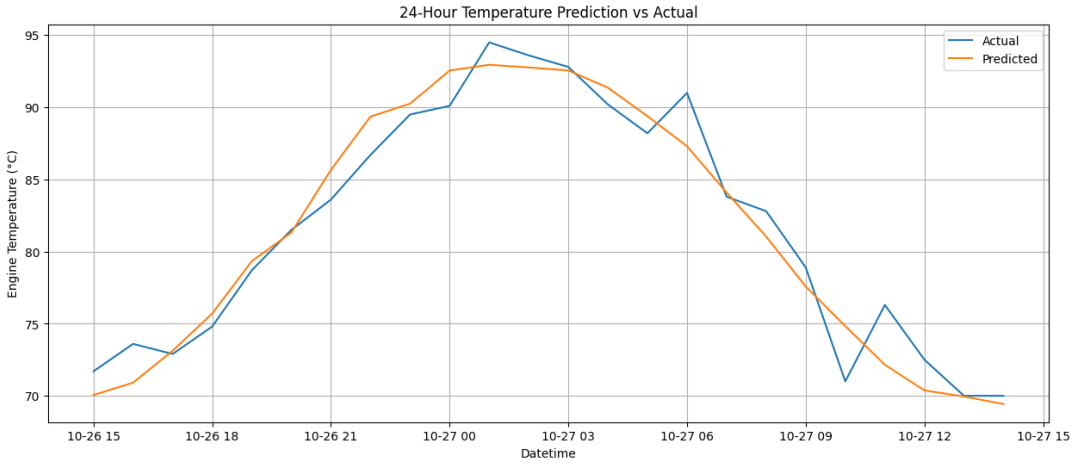


Figure 4-4 Histogram of Prediction Error Distribution. The plot shows the frequency of the model's prediction errors, which are centered around zero and approximate a normal distribution.

These results were validated on a test set composed of historical 2024 data that the models had not seen during training.

To further ensure robustness and generalizability, the models were also tested using synthetic 2025 data. This synthetic data was designed to simulate future operational conditions and include various edge cases and extreme value ranges (e.g., engine temperatures of 69.84-92.08°C, tyre pressures of 31.60-32.96 MPa). The models maintained their high accuracy on this synthetic data, confirming their ability to perform reliably in dynamic and evolving operational environments. The training process itself was optimized using an **Adam optimizer** and **Mean Squared Error (MSE)** as the loss function, with

early stopping implemented to prevent overfitting by halting training if the validation loss did not improve for a set number of epochs.

4.6 Deployment of Models and Integration with Swift Platform

A trained machine learning model is only useful if it can be deployed to make predictions on live data. The AI/ML subsystem is designed for a scalable and low-latency deployment.

1. **Model and Scaler Persistence:** The three trained Keras models are saved as .h5 files, and the corresponding MinMaxScaler objects used for data normalization are saved as .pkl files (e.g., scaler_TP.pkl) using Python's pickle library. Persisting the scalers is crucial to ensure that incoming live data is preprocessed in exactly the same way as the training data.
2. **API-based Deployment:** The models are deployed using FastAPI, a modern, high-performance Python web framework. FastAPI is used to wrap the models in API endpoints. This allows other services, such as the Swift Platform, to send live sensor data to the models via a simple HTTP request and receive predictions in return. This approach supports low-latency predictions (reported under 1 second), making it suitable for real-time decision-making.
3. **Integration Workflow:**

- ❖ The Swift Platform receives live sensor data from the on-vehicle hardware.
- ❖ For a given vehicle, Swift forwards a sequence of the recent historical data (e.g., the last 40-56 hours) in a JSON format to the appropriate FastAPI endpoint.
- ❖ The FastAPI application receives the data, preprocesses it using the loaded scaler, feeds it to the loaded .h5 model, and generates a forecast (e.g., an 8-hour pressure forecast).
- ❖ The model's predictions are inverse-scaled to their original units and sent back to the Swift Platform.
- ❖ The Swift Platform can then visualize these future trends on the dashboard or trigger a "predictive alert" if the forecasted values cross a predefined warning threshold.

This architecture ensures a seamless integration where the predictive power of the AI/ML models enhances the monitoring and alerting capabilities of the central Swift Platform.

4.7 Challenges in AI/ML Development

The development of the AI/ML subsystem involved navigating several common challenges inherent in machine learning projects:

- **Data Quality and Availability:** While the project started with clean datasets, ensuring a continuous supply of high-quality data from real-world sensors is a persistent challenge. Sensor noise, transmission errors, or downtime can impact data quality and require robust data validation and cleaning pipelines.
- **Feature Engineering Complexity:** Identifying the most impactful features is an iterative process. Determining the optimal lag periods, rolling window sizes, and relevant cyclical features required experimentation and domain knowledge to capture the underlying patterns effectively without introducing noise.
- **Model Tuning and Overfitting:** Training deep learning models like Bi-LSTMs involves tuning numerous hyperparameters (e.g., number of layers, units per layer, dropout rate, learning rate). Finding the optimal combination that yields the best performance without overfitting to the training data is a significant challenge. Techniques like Dropout, Batch Normalization, and Early Stopping were crucial in mitigating the risk of overfitting.
- **Computational Resources:** Training deep learning models on large time-series datasets can be computationally intensive, requiring significant time and processing power (CPU/GPU).
- **Validation and Generalization:** Ensuring that the models generalize well to new, unseen data and do not just memorize the training set is paramount. This was addressed through a rigorous testing strategy using a held-out test set and, importantly, testing against diverse synthetic data representing future and edge-case scenarios.

4.8 Summary of AI/ML Module

The AI/ML for Predictive Maintenance module represents the intelligent core of this project, transforming raw vehicle data into forward-looking, actionable insights. By leveraging sophisticated Bidirectional LSTM networks, the system successfully forecasts critical health parameters—engine temperature, tyre pressure, and tyre temperature—with a high degree of accuracy, as evidenced by test MAEs of 0.1795°C, 0.2368 MPa, and 0.344°C, respectively.

The success of this module is built upon a comprehensive development pipeline that includes meticulous data preprocessing, advanced feature engineering to capture complex temporal patterns, and a robust model architecture designed to prevent overfitting and ensure generalization. The models are validated against both historical and synthetic data, confirming their reliability for real-world deployment.

Deployed via a low-latency FastAPI framework, this AI/ML subsystem seamlessly integrates with the Swift Platform, providing the predictive capabilities necessary to shift from a reactive to a proactive maintenance strategy. It enables the system to anticipate failures before they occur, forming the foundation for reducing vehicle downtime, enhancing operational efficiency, and improving the safety of the flight catering fleet.

Chapter 5 ERP Integration with Salesforce

5.1 Introduction to ERP Integration and Maintenance Management

While the hardware and AI/ML modules provide the critical functions of data acquisition and predictive analysis, the true operational value of these insights is realized only when they are integrated into a structured and actionable workflow. In any industrial operation, including flight catering, maintenance is a core business process. This process involves identifying faults, scheduling technicians, managing spare parts, documenting work, and tracking costs. An

Enterprise Resource Planning (ERP) system is designed to manage and integrate these core processes.

This chapter details the third major technical component of the system: the integration with an ERP platform to automate and streamline the maintenance management workflow. For this project, Salesforce was selected as the ERP and service management platform. The integration is designed to create a seamless, closed-loop system where predictive alerts generated by the IoT and AI/ML subsystems are automatically translated into manageable maintenance tasks. This eliminates manual intervention, reduces response times, and provides a comprehensive, auditable record of all maintenance activities, thereby connecting predictive insights directly to operational execution.

5.2 Role of Salesforce in the Predictive Maintenance Workflow

1. Overview of Salesforce Service Cloud

Salesforce Service Cloud provides a suite of tools designed to automate service processes and make support and maintenance teams more efficient and effective. Key functionalities relevant to this project include:

- **Case Management:** At its core, Service Cloud allows for the creation and management of "Cases," which are records used to track any customer question, feedback, or issue. In this project, each alert or predicted failure from a vehicle is treated as a Case.
- **Workflow Automation**: Salesforce offers powerful declarative (clicks-not-code) automation tools like Flow and Process Builder. These tools can be used to define rules that automate business processes. For instance, an incoming API call from an external system can trigger a flow that automatically creates a new Case, assigns it to the correct maintenance team, and sends out email notifications.
- **Omni-Channel Routing**: Cases can be automatically routed to the right agent or queue based on predefined criteria, such as the type of fault, vehicle model, or technician skill set, ensuring the most qualified person handles the issue.
- **Knowledge Base**: Service Cloud includes a knowledge management feature that allows teams to build a repository of articles and standard operating procedures for resolving common issues, which can be attached to cases to guide technicians.

2. Ticket/Case Management for Maintenance

The core function of the Salesforce integration in this project is ticket management. As described by the user, the workflow is designed as follows:

- **Ticket Creation**: When the Swift Platform generates a critical alert (either from a direct sensor threshold breach or an AI/ML prediction), it triggers the creation of a new Case in Salesforce. This Case acts as a formal maintenance ticket.
- **Ticket Details**: The Case record contains all relevant information about the alert, such as the vehicle ID, the type of alert (e.g., "Tyre Pressure Low," "Engine Temperature High"), the severity, the timestamp, and the specific sensor readings that triggered the event.
- **Assignment and Action**: The Case is automatically assigned to the maintenance team. A technician can then view the ticket, understand the predicted issue, and schedule the necessary inspection or maintenance task before the vehicle is due for its next operation.
- **Status Tracking**: The technician updates the status of the Case within Salesforce as they work on it (e.g., from "New" to "In Progress" to "Resolved").
- **Closed-Loop Feedback**: Once the maintenance is complete and the Case is closed in Salesforce, this status change is synchronized back to the Swift Platform, clearing the alert and updating the vehicle's overall health status on the main dashboard.

This process transforms a raw data alert into a structured, trackable, and manageable work item, ensuring accountability and providing a complete audit trail for every maintenance action.

5.3 Integration Architecture: Swift Platform and Salesforce

The integration between the Swift Platform and Salesforce is designed as a bidirectional data exchange, facilitated primarily through APIs. This architecture ensures that data flows seamlessly between the operational monitoring system (Swift) and the maintenance management system (Salesforce).

1. **Data Flow from Swift to Salesforce (Alerts to Tickets)**

This is the primary forward-facing data flow that initiates the maintenance process.

- **Alert Trigger**: An alert is generated within the Swift Platform. This can be a simple rule-based alert (e.g., engine temperature > 80°C) or a more complex predictive alert from the AI/ML models (e.g., "tyre pressure is predicted to drop below a safe threshold within 8 hours").
- **API Call**: The Swift Platform's backend logic is configured to make a REST API call to a specific Salesforce API endpoint upon the trigger of a critical alert.
- **Case Creation:** This API call carries a payload (likely in JSON format) containing all the necessary information about the alert. Salesforce receives this request and, through its automation logic, creates a new Case record with the fields populated from the API payload.

2. **Data Flow from Salesforce to Swift (Status Updates)**

This is the feedback loop that closes the maintenance process and keeps the operational dashboard updated.

- **Status Change in Salesforce:** A maintenance technician resolves the issue with the vehicle and updates the status of the corresponding Case in Salesforce to "Closed" or "Resolved".
- **Outbound Trigger:** This status change can trigger an outbound action in Salesforce. This is typically configured using a Salesforce Flow or Apex trigger that initiates an outbound message or a direct API callout to a webhook endpoint on the Swift Platform.
- **Update in Swift Platform:** The Swift Platform receives this notification from Salesforce. It then updates its internal records to reflect that the maintenance has been completed, clears the corresponding alert from the active alerts list on the dashboard, and updates the vehicle's status back to "Active" or "Healthy".

5.4 Implementation Details

1. **Salesforce Configuration**

To support this workflow, the Salesforce environment was configured with the following elements:

- **Case Object Customization:** The standard "Case" object was likely

customized with additional fields to store vehicle-specific information, such as `Vehicle_ID`, `Alert_Type`, `Alert_Severity`, `Timestamp`, and a field to store the raw sensor data or a link back to the specific reading in Swift.

- **Automation Rules:** Salesforce Flow was used to build the core automation logic. A specific flow would be designed to:

  - Listen for incoming API requests on a dedicated endpoint.
  - Parse the JSON payload from the Swift Platform.
  - Create a new Case record and map the data from the payload to the corresponding Case fields.
  - Use assignment rules to automatically route the new Case to the appropriate maintenance queue or user.

- **Outbound Logic:** Another Flow or an Apex trigger would be configured to fire when a Case's status is changed to "Closed". This process would be responsible for constructing and sending the API call back to the Swift Platform to signal the resolution of the issue.

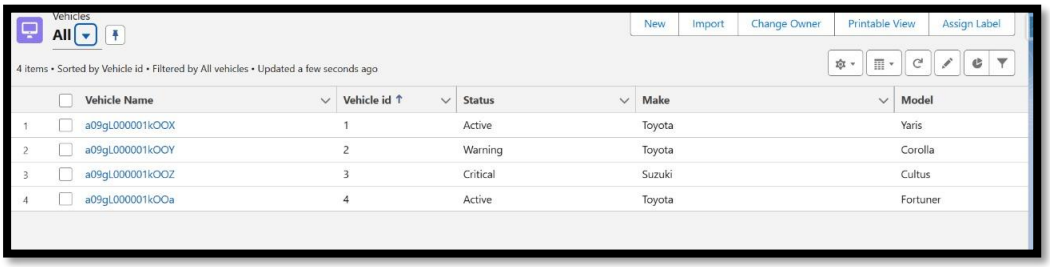


Figure 5-1 The 'Vehicles' list view in Salesforce, serving as the master record for each asset. Note the 'Status' column, which is updated in real-time based on system alerts

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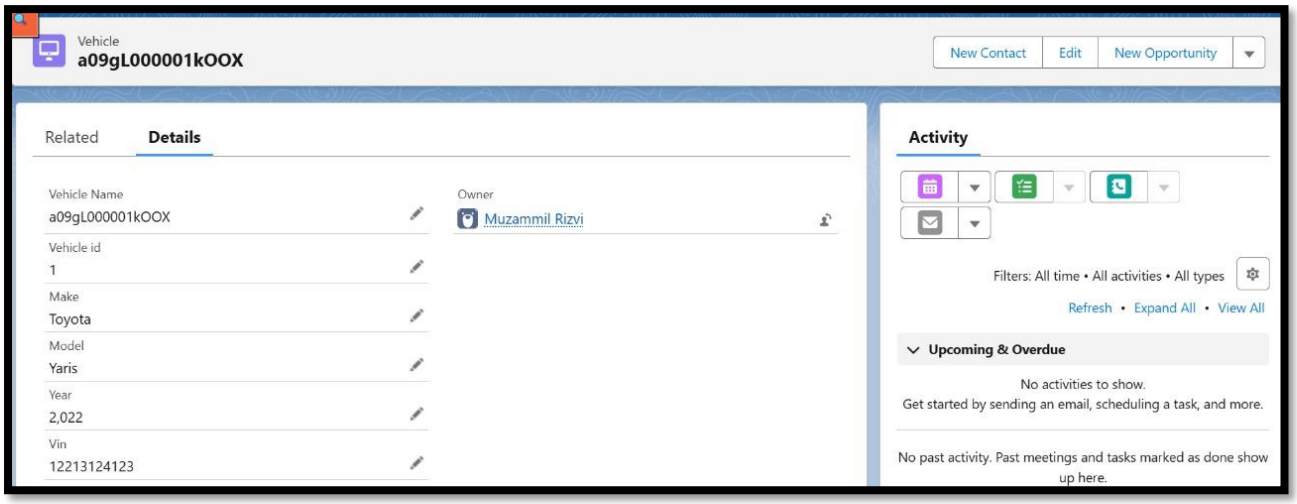


Figure 5-2 The detail page of a 'Vehicle' record, displaying its static attributes like Make, Model, and VIN.

2. API Integration

The integration relies on Salesforce's powerful and well-documented APIs.

- **Salesforce REST API**: The Swift Platform acts as a client that authenticates with Salesforce (likely via OAuth 2.0) and uses the standard sObject REST API to create records. A POST request to the /services/data/vXX.X/sobjects/Case/ endpoint, with the ticket details in the request body, is used to create a new Case.
- **Swift Platform Webhook:** The Swift Platform would expose a secure webhook (a custom API endpoint) to receive status updates from Salesforce. This webhook is designed to accept a payload containing the vehicle ID and the new status, allowing it to update the operational dashboard accordingly.

5.5 Use Case: Predictive Maintenance Ticket Lifecycle

To illustrate the end-to-end process, consider the following use case:

- **Prediction:** The AI/ML model forecasts that the pressure in the front-left tyre of Vehicle-10 is likely to fall below the critical safety threshold within the next 8 hours.
- **Alert & Ticket Creation:** The Swift Platform receives this prediction, flags it as a "Predictive Alert," and makes a REST API call to Salesforce. A new Case is automatically created in Salesforce with the subject "Predictive Alert: Low Tyre Pressure," linked to Vehicle-10.

- **Assignment:** The Case is automatically assigned to the "Tyre Maintenance Team" queue in Salesforce. Team members are notified.
- **Action:** A technician from the team picks up the Case. They locate Vehicle-10 using its last known location from the Swift dashboard and perform an inspection. They find a slow leak caused by a nail and repair the tyre.
- **Resolution:** The technician updates the Case in Salesforce, adding notes about the repair and changing the status to "Resolved."
- Synchronization: The status change in Salesforce triggers an outbound API call to the Swift Platform. The Swift dashboard updates, clearing the alert for Vehicle-10 and showing its status as healthy.
- **Record:** A permanent record of the predictive alert, the fault found, and the corrective action taken now exists in Salesforce, linked to Vehicle-10's maintenance history.

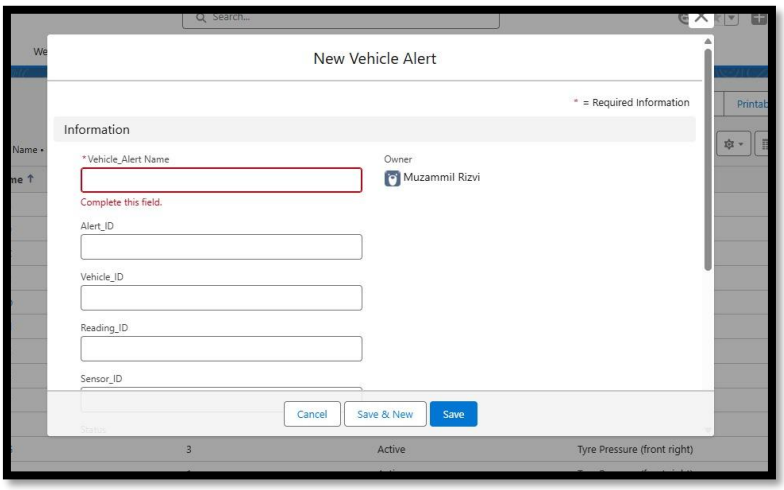


Figure 5-3 The 'New Vehicle Alert' creation interface, showing the fields that are automatically populated by the API call from the Swift Platform when an issue is detected.

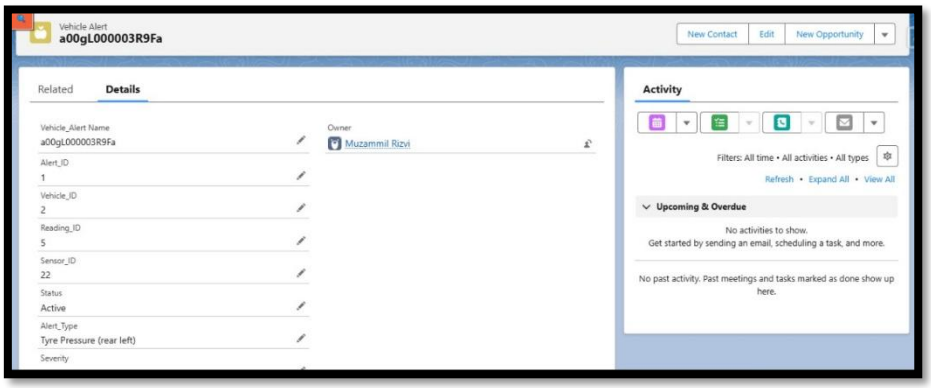


Figure 5-4 A detailed view of a 'Vehicle Alert' record in Salesforce. This record functions as the maintenance ticket, capturing all relevant data from the sensor event, such as the Vehicle ID, Sensor ID, and Alert Type.

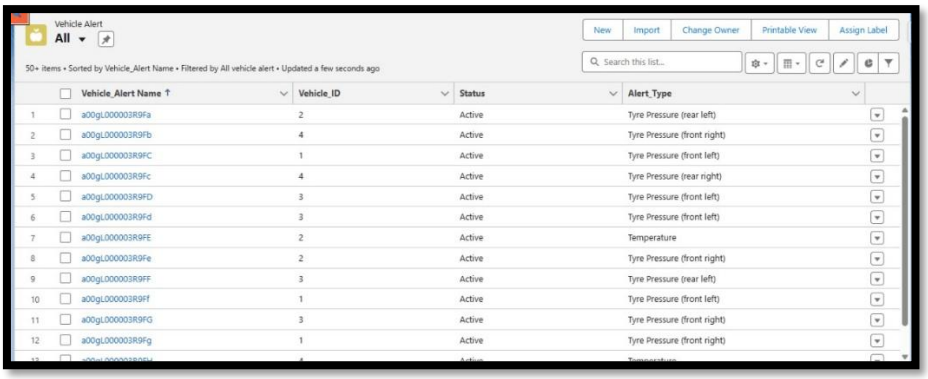


Figure 5-5 The 'Vehicle Alert' list view, which provides maintenance managers with a comprehensive overview of all active and historical maintenance tickets across the fleet.

5.6 Benefits of Salesforce Integration in the System

Integrating with Salesforce provides numerous benefits that elevate the system from a simple monitoring tool to a comprehensive management solution:

- **Streamlined Workflow**: Automates the entire process from fault detection to resolution, reducing manual effort and potential for human error.
- **Improved Response Time:** Immediately notifies the maintenance team when an issue is detected, allowing for faster scheduling and intervention.
- **Centralized Record Keeping**: Creates a single source of truth for all maintenance activities, providing a complete and auditable history for each vehicle.
- **Enhanced Accountability**: Clearly assigns ownership of maintenance tasks and tracks them through to completion.
- **Data-Driven Maintenance Strategy:** The historical data collected in Salesforce on faults and repairs can be analyzed to identify trends, common failure points, and areas for process improvement.
- **Scalability:** Leverages a robust, scalable enterprise platform designed to handle large volumes of data and complex business processes.

5.7 Challenges in ERP Integration

Integrating two complex platforms like Swift and Salesforce can present several challenges:

- **API Limits and Governance**: Salesforce has API request limits that

depend on the edition and user licenses. The integration must be designed efficiently to avoid exceeding these limits, especially if the fleet is large and generates many alerts.

- **Data Mapping and Transformation**: Ensuring that data is correctly mapped between the two systems is critical. Discrepancies in data formats or field types can lead to integration errors.
- **Authentication and Security**: Securely managing authentication (e.g., OAuth tokens) and ensuring that data is transmitted over secure channels (HTTPS) is paramount to protect sensitive operational data.
- **Error Handling:** A robust error-handling mechanism is required to manage situations where an API call fails, such as network issues or a temporary platform outage. The system needs logic to retry failed requests or notify administrators of the failure.
- **Synchronization Complexity**: Maintaining state and ensuring data consistency between two separate systems, especially in a bidirectional flow, can be complex to implement and maintain.

5.8 Summary of ERP Integration

The integration with Salesforce is a cornerstone of this project, transforming predictive insights into tangible maintenance actions. By leveraging Salesforce Service Cloud, the system automates the creation, assignment, and tracking of maintenance tickets triggered by real-time alerts. This bidirectional integration creates a closed-loop workflow where faults detected by the IoT/AI system are managed within a robust ERP environment, and resolution statuses are synchronized back to the operational dashboard. This process not only streamlines maintenance operations and improves efficiency but also builds a valuable historical dataset for future analysis. Despite challenges related to API management and data synchronization, the successful integration of Salesforce provides a scalable and powerful framework for managing the entire predictive maintenance lifecycle.

Chapter 6 System Implementation, Testing, and Overall Platform

6.1 Consolidated System Architecture (End-to-End Flow)

The "Smart Asset Tracking and Predictive Maintenance System" is implemented as a multi-layered, integrated architecture that seamlessly connects on-vehicle hardware with cloud-based platforms for analytics and workflow management. Each layer of the architecture performs a distinct function, creating a comprehensive data flow from sensor-level data acquisition to executive-level maintenance management.

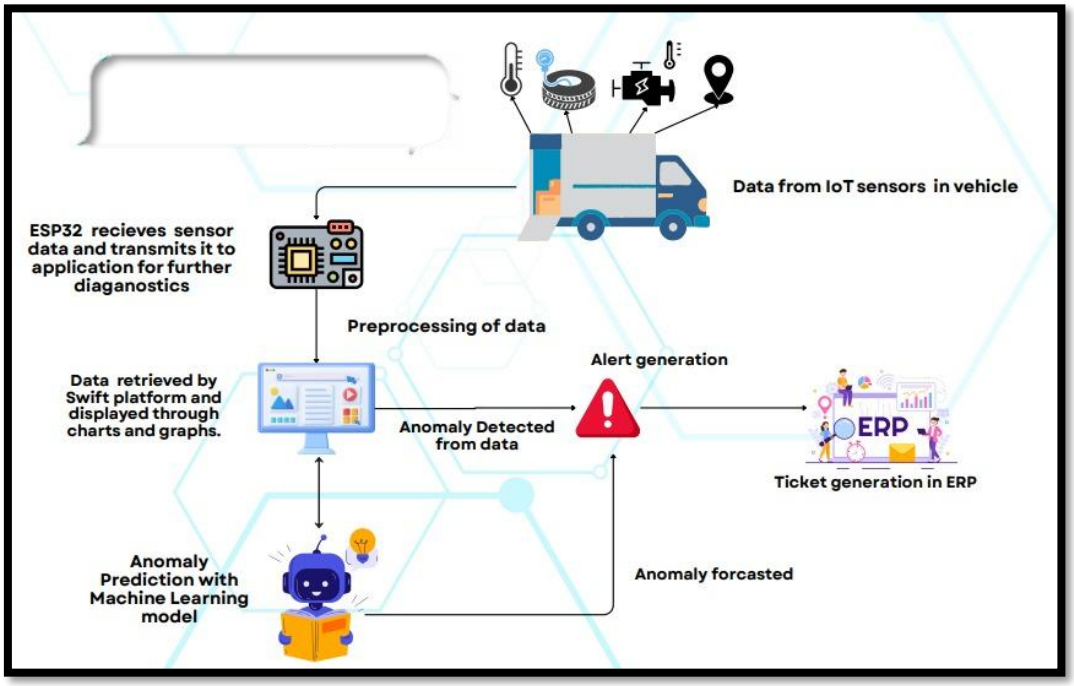


Figure 6-1 Consolidated System Architecture and End-to-End Data Flow.

The end-to-end data and process flow can be summarized in the following stages:

- **Data Acquisition (Hardware Layer):** The process begins at the vehicle with the custom-designed IoT unit. The ESP32-S3 microcontroller collects real-time data from the integrated sensor suite: location and speed from the NEO-6M GPS, engine temperature from the DS18B20 sensor, and tyre pressure/temperature from wireless BLE TPMS sensors. This architecture is visually represented in the hardware design flow diagram from the project's mid-year report.
- **Data Transmission & Aggregation (Platform Layer**): The ESP32

firmware formats the collected sensor data into JSON payloads and transmits it via Wi-Fi to the Swift Platform by Ephlux. The Swift Platform acts as the central data aggregation hub, receiving and storing the time-series data from the entire fleet of equipped vehicles.

- **Real-Time Monitoring & Basic Alerting (Platform Layer**): Fleet managers interact with the Swift Platform's dashboard to monitor the fleet in real-time. The platform visualizes vehicle locations on a map and displays current sensor readings. Simple rule-based alerts are also configured at this layer, where immediate threshold breaches (e.g., engine temperature exceeding 80°C) are flagged and displayed.
- **Predictive Analysis (AI/ML Layer**): Historical and real-time data stored on the Swift Platform is fed into the predictive analytics engine. The trained Bi-LSTM models process this data to forecast future values for engine temperature, tyre pressure, and tyre temperature. If a model predicts that a parameter will cross a critical threshold within its forecast horizon (e.g., 8 or 24 hours), it generates a "predictive alert."
- **Automated Maintenance Workflow (ERP Layer**): Both real-time and predictive alerts trigger the next stage of the workflow. The Swift Platform, upon flagging a critical alert, makes an API call to Salesforce. This automatically creates a new maintenance Case (ticket) in Salesforce Service Cloud, populated with details of the alert, vehicle, and sensor readings.
- **Maintenance Execution & Feedback Loop (ERP & Platform Layers):** The maintenance team manages the entire repair lifecycle within Salesforce. A technician is assigned the Case, performs the necessary maintenance, and updates the Case status to "Resolved." This status change in Salesforce triggers a reverse API call (a feedback loop) to the Swift Platform, which then updates the vehicle's status on the dashboard, clearing the alert and confirming the resolution. This end-to-end flow is depicted conceptually in the design flow diagram.

This consolidated architecture ensures that data flows logically from the point of creation to the point of action, creating an intelligent and automated system for fleet management.

6.2 Swift Platform: Central Hub and User Interface

The Swift Platform by Ephlux is a critical component of the system, serving

as the central integration hub and the primary user interface for fleet managers. It connects the hardware, the AI/ML models, and the ERP system.

1. **Dashboard Design for Data Visualization**

A key deliverable of the project is an intuitive and comprehensive dashboard developed on the Swift Platform. This dashboard provides a centralized view of the entire fleet's operations and health status. Based on the designs from the project's mid-year report, the dashboard includes several key widgets:

- **Map Visualization:** A real-time map, integrated using the Google Maps API, displays the current location of all vehicles. Vehicle markers can be color-coded to represent their status (e.g., green for active, orange for warning, red for critical), providing an at-a-glance overview of the fleet.
- **Vehicles Table:** A dynamic table lists all vehicles and their key real-time metrics, such as speed, engine temperature, and fuel level.
- **Detailed Analytics Graphs:** Users can select individual vehicles to view historical trends for specific sensor readings, such as fuel consumption over time or temperature fluctuations, presented as line graphs.
- 6.2.2 Real-time Alerting and Notification Features
- The Swift Platform serves as the primary interface for managing and viewing alerts.
- **Alerts Table:** A dedicated table on the dashboard highlights all active alerts, providing details such as the time of occurrence, the vehicle affected, the type of alert (e.g., "Tyre Pressure," "Engine Temperature"), and its severity.
- **Alerts Chart:** A visual chart, such as a bar chart, provides a summary of alerts by type, allowing managers to quickly identify common issues across the fleet.
- **Webhook Integration:** The platform's ability to receive data via webhooks from both the hardware and Salesforce is central to its function as a hub. It receives sensor data and alerts from the ESP32 and status updates from Salesforce, ensuring the information displayed on the dashboard is always current.

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6.3 System Testing and Validation

To ensure the reliability, accuracy, and robustness of the entire system, a multi-stage testing methodology was employed, covering individual components (unit testing), their interactions (integration testing), and the end-to-end performance of the system (system validation).

1. **Unit Testing**

Each major component of the system was tested in isolation to verify its functionality.

- **Hardware Unit Testing:** The ESP32-based hardware unit underwent functional validation to confirm that each sensor (GPS, DS18B20, TPMS) was correctly collecting and processing data. This included verifying data transmission protocols and ensuring that the JSON payloads were correctly formatted. Environmental robustness tests were also conducted, exposing the hardware to vibrations and a temperature range of 0-50°C to ensure its durability in real-world conditions.
- **AI/ML Model Unit Testing**: Each of the three Bi-LSTM models was individually tested on a held-out test dataset (from the 2024 data) to

evaluate its predictive accuracy. The performance was measured using MAE, and the results were compared against the target of <0.5 units.

- **ERP Trigger Unit Testing:** The automation rules (Flows) within Salesforce were tested in a sandbox environment. This involved simulating incoming API calls to verify that Cases were created correctly and that outbound messages were triggered appropriately upon status changes.

2. Integration Testing

This phase focused on testing the data flow and interactions between the major components.

- **Hardware-to-Platform Integration**: Tests were conducted to ensure that data transmitted from the ESP32 unit was being correctly received, parsed, and stored by the Swift Platform. This validated the entire data pipeline from the vehicle to the cloud.
- **Platform-to-AI Integration**: The process of feeding data from Swift to the deployed FastAPI model endpoints and receiving predictions back was tested to ensure seamless communication and correct data handling.
- **Platform-to-ERP Integration:** The alert-to-ticket creation process was tested end-to-end. Critical alerts generated in Swift were verified to automatically and accurately create Cases in Salesforce. The feedback loop, where a "Closed" Case in Salesforce correctly updated the status in Swift, was also validated.

3. System Validation

The final stage of testing involved validating the entire, consolidated system against realistic use cases.

- **Synthetic Data Testing:** The entire system was tested using synthetic 2025 data profiles to validate its performance under diverse and future operational scenarios, including edge cases with extreme sensor values.

This confirmed the robustness of both the hardware's data handling and the AI models' predictive accuracy under conditions not present in the historical training data. For example, the models' predictions remained robust with MAEs below the 0.5 threshold even when tested with synthetic temperature ranges of 69.84-92.08°C.

- **End-to-End Scenario Testing:** The complete workflow, from a simulated sensor anomaly on the vehicle to the closure of the corresponding maintenance ticket in Salesforce and the final status update on the Swift dashboard, was tested to ensure all components worked together as expected.

6.4 **Test Results and Evaluation**

The system demonstrated strong performance and reliability across all stages of testing.

- **Hardware Performance:** The hardware unit performed reliably during environmental tests, and data transmission latency was low, with sensor payloads being sent in under 1 second, supporting near real-time monitoring.
- **AI/ML Model Accuracy:** As detailed in Chapter 4, the models achieved excellent predictive accuracy on the test data, with MAEs of 0.1795°C for engine temperature, 0.2368 MPa for tyre pressure, and 0.344°C for tyre temperature. This high level of accuracy ensures that predictive alerts are reliable and minimizes the risk of false positives.
- **Integration Reliability**: The integration tests confirmed that data flowed reliably between all system components. The automated ticket creation in Salesforce from Swift alerts was successful, with an average latency of a few seconds from alert generation to Case creation.

Overall, the evaluation results confirm that the system meets its design objectives for accuracy, reliability, and functionality. The successful integration of all components validates the project's architecture and its capability to function as an effective predictive maintenance solution.

6.5 Challenges Encountered During Implementation and Integration

The implementation of a complex, multi-platform system presented several technical challenges that required careful consideration and problem-solving:

- **Hardware and Firmware**: Ensuring stable BLE connectivity for the TPMS sensors in an electronically noisy environment was a challenge. Fine-tuning the scan parameters and ensuring robust parsing logic for advertisement packets was an iterative process. Managing the concurrent tasks of sensor polling, BLE scanning, and Wi-Fi transmission in the ESP32 firmware without causing blocking or instability required careful code structuring.
- **AI/ML Development:** A significant challenge was preventing the deep learning models from overfitting. This was addressed by implementing regularization techniques like Dropout and Early Stopping, as well as by validating the models against diverse synthetic datasets.
- **Platform Integration**: The primary challenge was ensuring seamless and error-free data synchronization between three distinct platforms (Hardware/ESP32, Swift, and Salesforce). This involved debugging API calls, managing authentication tokens securely, and implementing robust error-handling and retry logic for cases where an API call might fail due to network issues. Mapping data fields correctly between the Swift alert payload and the Salesforce Case object required meticulous attention to detail.

6.6 Discussion of Results

The successful implementation and positive test results of the "Smart Asset Tracking and Predictive Maintenance System" demonstrate the significant potential of integrating IoT, AI/ML, and ERP technologies to solve tangible problems in industrial fleet management. The high accuracy of the AI/ML models confirms that a data-driven, predictive approach is superior to traditional reactive or scheduled maintenance. The ability to forecast potential failures up to 24 hours in advance provides a practical and valuable window for maintenance teams to act proactively.

The seamless integration with Salesforce is a key achievement, as it closes the loop between insight and action. It ensures that every predictive alert is converted into a trackable, manageable task, thereby embedding the predictive

maintenance process into the organization's standard operational workflow. This not only enhances efficiency but also provides a rich source of data within Salesforce that can be used for long-term analysis of vehicle reliability, maintenance effectiveness, and component lifespan. The final system effectively proves that the project's initial aim of creating a comprehensive, end-to-end solution is achievable and provides a strong foundation for future development and deployment.

6.7 Summary

This chapter has detailed the implementation of the consolidated system, highlighting how the hardware, AI/ML, and ERP components function as a cohesive whole. The architecture facilitates a logical end-to-end data flow, from real-time sensor acquisition at the vehicle to automated maintenance ticket management in Salesforce, with the Swift Platform acting as the central monitoring and integration hub. A rigorous, multi-stage testing methodology validated the functionality, accuracy, and reliability of each component and their integrations. The results confirm that the system successfully meets its objectives, providing a robust and effective platform for smart asset tracking and predictive maintenance. The challenges encountered during implementation were systematically addressed, leading to a mature and well-tested proof-of-concept system ready for further development.

Chapter 7 Conclusion and Future Work

7.1 Conclusion

This project successfully designed, developed, and validated the "Smart Asset Tracking and Predictive Maintenance System for Flight Catering," a comprehensive solution addressing critical inefficiencies in industrial fleet management. The system effectively integrates custom Internet of Things (IoT) hardware, advanced Artificial Intelligence (AI) and Machine Learning (ML) models, and an Enterprise Resource Planning (ERP) platform to create a cohesive, end-to-end ecosystem. By shifting the maintenance paradigm from a reactive to a proactive, data-driven approach, the project has demonstrated a clear pathway to enhancing operational reliability, reducing costs, and improving safety in the demanding flight catering industry.

The project successfully met its core objectives:

- **A robust hardware unit** was developed using an ESP32-S3 microcontroller, capable of acquiring real-time data from GPS, temperature (DS18B20), and BLE TPMS sensors, and reliably transmitting it to the cloud.
- **High-accuracy predictive maintenance models** were created using Bi-LSTM networks. These models achieved exceptional performance, with test Mean Absolute Errors (MAEs) of 0.1795°C for engine temperature, 0.2368 MPa for tyre pressure, and 0.344°C for tyre temperature, validating their effectiveness in forecasting component health.
- **A seamless ERP integration** with Salesforce was designed and implemented, automating the creation of maintenance tickets from system alerts and establishing a closed-loop workflow for managing repairs and updating system status.
- **A centralized monitoring platform** was conceptualized and implemented on the Swift Platform, providing an intuitive dashboard for real-time vehicle tracking, data visualization, and alert management.

The system as a whole was validated through rigorous testing, confirming its functional and operational integrity. The project serves as a powerful proof-of-concept, showcasing how the strategic combination of modern technologies

can lead to significant improvements in traditional industrial processes, with an estimated potential to reduce fleet downtime and operating expenses by 10-15%.

7.2 Limitations of the Current System

Despite the successful implementation and validation, the current system has several limitations that should be acknowledged as areas for potential future improvement:

- **Sensor Dependency and Reliability:** The system's performance is fundamentally contingent upon the accuracy and reliability of the on-vehicle sensors. A malfunctioning sensor (e.g., TPMS, DS18B20) could lead to inaccurate data, potentially causing missed alerts or false positives. The current system has basic data validation but could benefit from more advanced algorithms to detect and handle anomalous readings caused by sensor faults.
- **Scope of Predicted Parameters:** The AI/ML models are currently focused on three critical health indicators (engine temperature, tyre pressure, tyre temperature). A more comprehensive vehicle health assessment would require monitoring and predicting failures in other subsystems, such as the transmission, electrical system, or brakes.
- **Data Security:** While standard security practices like encrypted Wi-Fi are used, the data transmission pipeline could be further hardened. The planned implementation of HTTPS for all API calls between the hardware and cloud platforms is a necessary next step to ensure end-to-end encryption.
- **Model Maintenance and Retraining:** The AI/ML models require periodic retraining with new operational data to maintain their accuracy as vehicle behavior evolves and new data patterns emerge. This implies an ongoing operational overhead for data management and model lifecycle management.
- **Contextual Factors:** The feature engineering process captures some operational context (e.g., rush hours), but other external variables that could influence vehicle health, such as specific cargo loads, road surface conditions, or individual driving styles, are not explicitly modeled in the current iteration.

7.3 Recommendations for Future Work

Building upon the successful foundation of this project, several enhancements could be pursued to further extend its capabilities and impact.

1. **AI/ML Enhancements:**

   - **Expansion of Predictive Capabilities**: Develop new models to predict failure modes for other critical vehicle components by incorporating a wider range of sensor data.
   - **Advanced Model Architectures:** Explore more recent neural network architectures, such as Transformers, which have shown great promise in time-series forecasting, to determine if they can offer further accuracy improvements over Bi-LSTMs.
   - **Explainable AI (XAI):** Integrate XAI techniques to provide more transparent and interpretable insights into why a model is predicting a potential failure. This would enhance user trust and provide more actionable information for maintenance technicians.
   - **Dynamic Anomaly Thresholds:** Investigate methods for dynamically adjusting alert thresholds based on vehicle type, age, or specific operating conditions to further reduce false positives and improve alert relevance.

2. **Hardware and System Enhancements:**

   - **Full CAN Bus Data Integration:** Actively integrate the MCP2515 CAN module already present on the PCB to read a wealth of additional diagnostic data directly from the vehicle's CAN bus (e.g., engine RPM, fault codes, fuel consumption). This would significantly enrich the feature set for existing AI/ML models and enable the prediction of new failure types.
   - **Additional Sensor Integration**: Incorporate other relevant sensors, such as accelerometers for vibration analysis (useful for detecting bearing or suspension issues) or current sensors for monitoring the electrical system.
   - **Enhanced Data Security:** Fully implement HTTPS for all data transmissions between the ESP32 and backend servers to ensure robust end-to-end encryption.

- **Edge Computing Capabilities**: Explore performing lightweight model inference or more complex data preprocessing directly on a more powerful edge device within the vehicle. This could reduce data transmission loads and enable faster alerts, especially in areas with poor connectivity.

3. **Platform and Integration Enhancements:**

- **Advanced Swift Platform Features**: Continue to develop the Swift platform's dashboard with more sophisticated data visualization tools, customizable alert functionalities, and multi-vehicle fleet management views.
- **Deeper ERP/Salesforce Integration**: Enhance the Salesforce integration to include spare parts inventory management linked to maintenance tickets and to generate more detailed analytics on maintenance costs and technician performance.

7.4 Final Remarks

The "Smart Asset Tracking and Predictive Maintenance System for Flight Catering" project successfully demonstrates the power of integrating modern digital technologies to revolutionize a traditional industrial process. By creating a system that not only monitors but also predicts and streamlines, this work provides a robust framework for enhancing efficiency, ensuring reliability, and driving down costs. The positive results achieved serve as a strong testament to the project's design and execution, offering a scalable and impactful solution poised to make a significant contribution to the operational excellence of the flight catering industry and other fleet-dependent sectors.

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